Robust Patch-Level Metastasis Detection Under Domain Shift: Preprocessing, Foundation-Model Representations, and Bidirectional External Validation

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Declaration

This is to declare that the graduation project entitled "Robust Patch-Level Metastasis Detection Under Domain Shift: Preprocessing, Foundation-Model Representations, and Bidirectional External Validation" under the supervision of Dr. Sameer Hasan is our own work and does not contain any unacknowledged work from any other source.

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Acknowledgement

أَوَّلًا وَقَبْلَ كُلِّ شَيْءٍ، نَحْمَدُ اللهَ سُبْحَانَهُ وَتَعَالَى وَنَشْكُرُهُ عَلَى تَوْفِيقِهِ، وَكَرَمِهِ، وَتَيْسِيرِهِ لَنَا لِإِتْمَامِ هَذَا الْعَمَلِ.

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Abstract

Sentinel lymph node assessment on hematoxylin and eosin (H&E) sections underpins breast cancer staging but is heavily limited by inter-observer variability and manual screening latency. While patch-level deep learning provides a scalable route to computational decision support, model performance often degrades under domain shift arising from institutional variations in staining, tissue preparation, and scanner hardware.

This thesis develops a rigorous, reproducible preprocessing and classification framework utilizing PatchCamelyon (PCam) and CAMELYON17 (WILDS) to evaluate model behaviour under cross-domain transfer. The principal external question is whether models trained on PCam generalize to a held-out multi-hospital CAMELYON17 test set (the stress case that motivated integrity-aware curation) while a prespecified mirror (train on CAMELYON17, test on PCam) is reported to characterize transfer asymmetry and is not treated as a symmetric second success criterion for that forward claim. A fixed data-integrity pipeline enforces strict split isolation, tissue-quality filtering, cross-corpus overlap screening, and single-reference Macenko stain normalization after a prespecified shallow-CNN benchmark. Model selection and post-hoc temperature scaling are strictly restricted to source-domain validation data. Two distinct learning architectures are compared under identical data constraints: a compact convolutional neural network (CNN) trained end-to-end, and Virchow2 configured with a frozen feature extractor and trainable linear head, thereby contrasting bottlenecks mitigated by quality-aware preprocessing with those addressed by large-scale representation, including how far quality-aware preprocessing remains necessary when using a frozen pathology foundation backbone compared with an end-to-end shallow CNN on the same tensors.

The empirical results separate structural preprocessing effects from representation capacity. On raw, un-normalized inputs, the baseline CNN trained on CAMELYON17 exhibited suboptimal in-domain performance, yielding an Area Under the Receiver Operating Characteristic curve (ROC-AUC) of 0.724 and a recall of 0.16 at a probability threshold of 0.5. Integrating the proposed preprocessing pipeline successfully recovered this capacity, raising the in-domain CAMELYON17 ROC-AUC to 0.985 and recall to 0.91 at the 0.5 threshold, with clear gains on PCam-related surfaces as well. Virchow2 consistently outperformed the shallow baseline on preprocessed patches across all experimental conditions, displaying the strongest practical margin on the forward PCam-to-CAMELYON17 external test.

Crucially, the prespecified mirror layout exposes a distinct transfer asymmetry that the headline forward path alone would obscure: while transfer from PCam to external CAMELYON17 remained highly robust for both architectures, CAMELYON17-trained models showed severe threshold and calibration degradation on external PCam. For Virchow2, external PCam ranking remained competitive with an ROC-AUC of 0.950, but recall at the default 0.5 threshold fell to 0.630, generating a substantial false-negative pool linked in structured qualitative review to small-focus lesions and context boundaries. Post-hoc temperature scaling improved probabilistic log loss but did not correct threshold-level bias; the expected calibration error (ECE_{15}) rose from 0.020 on in-domain PCam to 0.160 on external PCam after CAMELYON17 training. Bootstrap percentile intervals and paired permutation tests (with multiplicity control) supported these planned comparisons.

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Chapter 1 Introduction

1.1 Clinical and Histopathological Foundations

Histology is the microscopic study of the structure and composition of biological tissues and has been foundational to medical diagnosis since the nineteenth century [1] [2]. By preparing thin sections of tissue and staining them with dyes that selectively colour different cellular components, histologists render the invisible architecture of organs and tissues visible under a microscope [2].

Histopathology extends this discipline to the study of diseased tissues, serving as the gold standard for diagnosing a wide range of conditions, from infections to malignancies [3] [4]. When abnormal changes in cell morphology, tissue architecture, or staining patterns are identified, they provide crucial evidence for determining the nature, extent, and severity of disease.

Among the most consequential applications of histopathology is cancer diagnosis and staging. Cancer is the uncontrolled proliferation and spread of abnormal cells; it remains a leading cause of death worldwide [5]. A critical step in determining a patient's prognosis and treatment plan is assessing whether the tumour has metastasized, that is, disseminated beyond its primary site to distant organs and tissues, most commonly through the lymphatic system [6]. Lymph nodes, which function as filters of tissue fluids and as sites of lymphocyte production, are therefore routinely examined histopathologically to detect metastatic deposits [7]. In breast cancer, the histopathological assessment of sentinel lymph node biopsies directly underpins the TNM staging system and informs treatment decisions [8].

To perform these diagnostic assessments, standard histopathological examination relies fundamentally on microscopy and tissue staining. The light microscope, first assembled by Zacharias Jansen in the sixteenth century and perfected by Leeuwenhoek in 1673, remains the primary tool for routine diagnostic work, with resolution governed by the ratio between the wavelength of light and the width of the diaphragm opening [4]. More advanced instruments, including scanning electron microscopes (SEM), transmission electron microscopes (TEM), cryo-electron microscopes (Cryo-EM), fluorescence microscopes, and phase-contrast microscopes, extend resolution to the ultrastructural and even atomic level for specialized applications [4].

The most widely used stain in histopathology is hematoxylin and eosin (H&E), first described in 1875–1878, in which hematoxylin colours cell nuclei blue and eosin stains the cytoplasm and extracellular matrix pink, enabling pathologists to evaluate tissue architecture and cellular morphology [1] [9]. Immunohistochemistry (IHC) further advances diagnostic capability by using labelled antibodies to detect specific antigens within tissue sections, enabling the identification of particular cell types, tumour markers, and prognostic biomarkers such as HER2 amplification in breast cancer [10], [11]. However, the preparation of high-quality slides is itself a complex, multi-step process, involving fixation, processing, embedding, sectioning, staining, and mounting, each step of which can introduce artefacts that alter tissue morphology and potentially compromise diagnostic accuracy [3].

Beyond the physical preparation of the tissue, the subsequent manual examination of whole-slide images is time-consuming, subjective, and prone to inter-observer variability for instance, breast cancer subtyping by expert pathologists has shown accuracy rates as low as 57.08%, and prostate cancer grading inter-observer agreement has been reported at only 57.90% [12].

Given that cancer diagnosis depends on such subtle morphologic distinctions, histopathologic interpretation is usually performed by specialist pathologists, and second-opinion review is often warranted when the diagnosis may alter staging or treatment. In a large breast-biopsy study, overall agreement between individual pathologists and a consensus reference was 75.3%, with the weakest agreement in atypia and ductal carcinoma in situ [13]. Specialist review also amended 16.5% of pediatric cancer diagnoses and in early rectal cancer discordant histopathologic assessment changed treatment in 11% of patients [14] a, [15].

A separate review of anatomic pathology error detection reported that interpretive error rates can range from 1.2 to 50 errors per 1,000 cases, and it highlighted double-reading, case conferences, and consultation as the traditional quality-control triad [16].

1.2 Digital Pathology and Domain Challenges

While traditional quality-control relies on human consultation and manual glass-slide review, the integration of digital pathology has introduced a new technological layer to this workflow, allowing whole-slide images (WSIs) to be viewed on a computer screen for remote consultation, annotation, and computational analysis. Open-source tools used in this workflow include QuPath, HistoQC, and HistomicsTK [17]. However, moving glass slides to the computational domain introduces severe infrastructure bottlenecks [18]. The sheer size of WSIs, with the CAMELYON dataset alone comprising 1,399 WSIs and consuming roughly

2.95 terabytes of storage, prevents them from being loaded into standard GPU memory in their entirety [19], [20], [21]. This constraint has necessitated patch-based processing paradigms, in which WSIs are tiled into small patches (typically 96×96 or 256×256 pixels), classified individually, and then aggregated into slide-level or patient-level predictions [22], [23].

To ensure these patches are of sufficient quality, preprocessing steps, including tissue detection, background removal, and stain normalization, are essential [24]. A primary challenge in this stage is "domain shift," where model performance degrades on images from unseen institutions due to variations in staining protocols, reagent brands, tissue preparation, and scanner hardware. Resolving this institutional colour and textural variability is critical, yet the optimal interaction between explicit normalization pipelines and modern neural networks remains an active area of investigation.

1.3 Evolution of Computational Algorithms

Early computational approaches to histopathology relied on classical machine learning and manually engineered features, such as Local Binary Patterns (LBP) and Scale-Invariant Feature Transform (SIFT), classified using Support Vector Machines (SVMs) or Random Forests [25].

While these methods demonstrated the feasibility of automated detection in breast and colourectal cancers, they were heavily dependent on human intuition for feature selection and often failed to capture the full heterogeneity of tumour microenvironments [26]. When compared directly on breast cancer datasets, deep learning approaches achieved superior accuracies of 94–98% for binary classification, compared to 85–89% for classical methods [27]. This formal shift toward Convolutional Neural Networks (CNNs) allowed models to learn hierarchical feature representations directly from raw pixel data, eliminating the need for handcrafted engineering [28]. Progress was further accelerated by transfer learning, where fine-tuned architectures like VGG16 and ResNet50 significantly outperformed models trained from scratch [29] [30].

More recently, self-supervised learning and pathology-specific foundation models, including CTransPath, UNI, CONCH, GigaPath, and the Virchow family, pretrained on millions of whole-slide images, have pushed the state of the art further, achieving strong performance across diverse tasks with minimal labelled data. Despite these advances, current pipelines still face substantial bottlenecks regarding institutional stain variation, model miscalibration, and a distinct lack of rigorous external validation across disparate clinical centres.

1.4 Problem Statement

Sentinel lymph-node assessment on H&E underpins breast cancer staging, yet manual review is slow and variable across observers. Patch-based AI on whole-slide images could support more consistent screening, but performance often degrades when staining, scanners, or laboratory workflows differ between sites, a form of domain shift that classical stain normalization was designed to address. Modern pathology foundation models are pretrained on very large, heterogeneous slide collections and may already absorb much stain-related variation, so it is unclear whether careful, quality-aware preprocessing remains necessary for reliable cross-centre use. This project tests that question for patch-level lymph-node metastasis detection by comparing a reproducible preprocessing pipeline (quality control and Macenko stain normalization) with a Virchow2 classifier and a conventional CNN baseline, under bidirectional external validation between PatchCamelyon and CAMELYON17 and evaluation of discrimination, calibration, and structured qualitative error patterns, not only in-dataset accuracy.

1.4.1 Aims

This project aims to build and evaluate a reproducible patch-level framework for lymph-node metastasis on H&E images, using PatchCamelyon (PCam) and CAMELYON17 (WILDS) under their official splits. The main aim is strong external performance when training on PCam and testing on held-out CAMELYON17 test (multi-hospital nodal tissue that stresses stain and acquisition differences relative to PCam). Integrity checks, quality screening, stain normalization (after a prespecified benchmark), and validation-bound training and calibration are aligned with that forward target, while keeping both corpora on a common pipeline for fair comparison.

Virchow2 (frozen encoder, trainable head) is compared to a shallow CNN on identical tensors and reporting rules. A prespecified mirror (train on CAMELYON17, test on PCam) is run to describe transfer asymmetry (for example when ranking stays high but threshold or calibration behaviour shifts); it informs interpretation and is not treated as a second bar that must be met for the PCam→CAMELYON17 claim. Additional aims are to quantify how much preprocessing still matters beside a frozen foundation encoder, and to report calibration, uncertainty, and structured qualitative review next to discrimination.

1.4.2 Objectives

To achieve these aims, the project will:

- Define a clinically relevant patch-level task on H&E lymph-node tissue (metastasis present versus absent in the central region) and adopt PCam and CAMELYON17 (WILDS) with their official splits, documenting label meaning, patch size, and in-domain versus external-domain evaluation.
- Establish a unified, traceable preprocessing pipeline with fixed project standards, applicable across both datasets.
- Implement data-integrity controls including exact deduplication on PCam, quality screening on both corpora, cross-dataset overlap checks, and auditable manifests so that performance estimates are not inflated by duplicates or leakage.
- Characterize both datasets through exploratory and audit workflows (class balance, tissue and colour statistics, quality-filter impact, split-wise summaries) to support methodological transparency in the report.
- Compare stain-handling strategies under matched conditions, select a primary normalizer for full-corpus preprocessing, and document the rationale using prespecified benchmark criteria.
- Train and evaluate a conventional CNN baseline on PCam and CAMELYON17, including raw public patches and study-pipeline inputs, to quantify preprocessing effects and transfer behaviour before and alongside foundation-model evaluation.
- Develop Virchow2-based classifiers with a frozen encoder and trainable head, consistent input preparation, and reproducible experiment artefacts (weights, logs, and predictions).
- Treat PCam train → CAMELYON17 test as the primary external criterion (with in-domain PCam test as reference), reporting discrimination, calibration, and errors on the multi-hospital split.
- Run the mirror (CAMELYON17 train → PCam test) to document asymmetry and ranking versus threshold/calibration effects; use it for interpretation, not as a symmetric requirement for the forward claim.
- Assess predictive reliability using calibration-focused metrics and post-hoc temperature scaling fitted on validation only, separating ranking performance (ROC-AUC and PR-AUC) from probability quality.

- Stratify test errors using predictive confidence and entropy on deterministic outputs, alongside standard false positives and false negatives.
- Perform structured qualitative error analysis using predefined buckets, a human review checklist, and exported review packages for case-based interpretation.
- Document reproducibility and experimental governance through versioned scripts, run manifests, naming conventions for experimental conditions, and explicit separation of corpus curation from per-patch preprocessing so that the pipeline can be replicated or extended.

In sum, the study tests whether a PCam-trained classifier (integrity-aware pipeline; Virchow2 or CNN) performs strongly on external CAMELYON17 test, how large the CNN–Virchow2 gap is on that path, and what the mirror adds about asymmetric transfer. It also asks whether quality-aware preprocessing remains necessary when using a frozen pathology foundation encoder (Virchow2), compared with the same pipeline for a conventional CNN. The artefacts are meant to support later work on more corpora or sites and on clinical-style operating points, without claiming deployment from patch benchmarks alone.

Chapter 2 Theoretical Background

This chapter introduces the clinical, imaging, and machine-learning concepts used later in the literature review, methodology, and results. It is descriptive theory only: benchmark names, preprocessing choices, and evaluation protocols for the present study are specified in those chapters.

2.1 Breast Cancer and Clinical Context

Breast cancer represents a substantial healthcare burden globally, as shown statistically in Figure 2.1. Globally, it stands as the most prevalent malignancy among women and the second leading cause of cancer-related mortality in developed nations, accounting for approximately 44,000 deaths annually in the United States alone. The high prevalence and mortality rates underscore the critical need for accurate diagnostic and prognostic tools in clinical practice [31] [32].

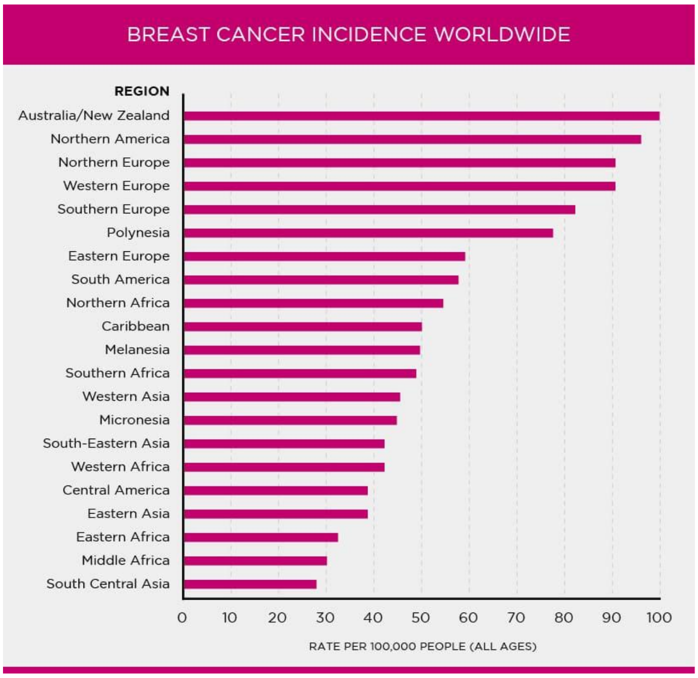

Figure 2.1. Breast Cancer statistics across several countries [33]

Breast carcinoma arises from the terminal duct lobular unit (TDLU) and is classified based on neoplasm infiltration patterns. The disease can be divided into invasive ductal carcinoma (IDC) and ductal carcinoma in situ (DCIS), with IDC representing the most common subtype [34]. Invasive breast cancer carries a poorer prognosis as it spreads from the original site, making precise tumour delineation crucial for further pathological analysis [32].

Metastatic presence in lymph nodes constitutes one of the most important prognostic variables in breast cancer [35]. Assessment of cancer spread to sentinel axillary lymph nodes (SLNs) is essential for accurate staging. Clinical practice differentiates between macrometastases, micrometastases, and isolated tumour cells, each carrying different clinical significance and treatment implications. However, the sensitivity of pathologists in SLN assessment is suboptimal, and the process is time-consuming, with micrometastases being particularly challenging to detect [36].

Histopathology plays a central role in breast cancer diagnosis and prognosis through microscopic analysis of tissue samples. Definitive diagnosis relies on biopsy examination, with manual identification of tumour presence and extent being critical for patient management, tumour staging, and assessing treatment response [37]. Histopathological tumour grading serves as a strong prognostic marker for breast cancer survival, as shown in Figure 2.2. These combine three key features: nuclear pleomorphism, tubule formation, and mitotic count. The histological grade is commonly determined according to the modified Bloom-Richardson system, which involves semi-quantitative assessment of nuclear atypia, tubule formation, and mitotic activity in H&E stained sections. Manual counting of mitotic tumour cells represents one of the strongest prognostic markers, though current methods are time-consuming and error-prone [38].

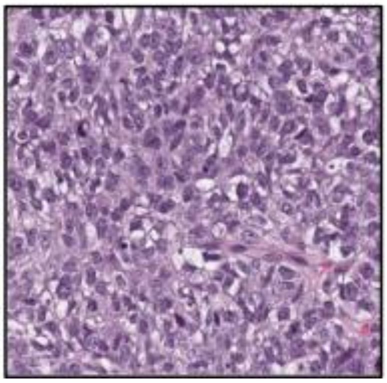

Figure 2.2. Histopathology slide of breast tissue exhibiting high tumour cellularity [39].

Nuclear morphology is key in cancer grading schemes, with nuclear arrangement and morphology correlating with patient outcomes. The morphology of tubules correlates with cancer aggressiveness, with later stages showing disorganization. Epithelium and stroma patterns are critical for predicting survival and outcome, with histologic patterns in stroma potentially influencing patient prognosis [34]. IHC further categorizes tumours using markers such as estrogen receptor (ER), progesterone receptor (PR), HER2, and Ki67. The analysis of immunohistochemically stained slides mainly involves estimating the number of cells positive for a particular antigen and the degree of positivity (staining intensity) [40].

Manual pathology is tedious and prone to high observer variability [37]. Reviewing sentinel lymph nodes is particularly time-consuming and subjective, as metastases can be extremely small or difficult for the human eye to detect [35]. Variations in staining and scanning across labs further necessitate computerized decision support tools. These methods can reduce workload and improve diagnostic accuracy by assisting pathologists in localizing breast cancer [32].

2.2 Histological Staining

2.2.1 H&E Stain Chemistry and Morphology Interpretation

Hematoxylin and eosin staining represents the standard protocol in histopathology, imparting a bluish colour to nuclei and a pinkish colour to the cytoplasm, as depicted in Figure 2.3. The H&E staining process follows a standard protocol, though the appearance of stained slides varies among pathology laboratories and even within the same centre over time. This staining technique is fundamental for visualising tissue morphology and enables pathologists to perform semi-quantitative assessment of nuclear atypia, tubule formation, and mitotic activity [41].

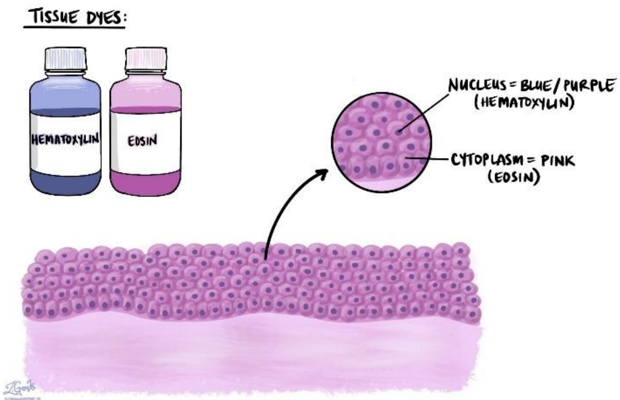

Figure 2.3. Hematoxylin and Eosin (H&E) Staining in Histopathology [42].

The Lambert-Beer law describes the relationship between light absorption and stain amount, serving as a theoretical foundation for understanding stain chemistry. Optical density (OD) vectors can be quantified for hematoxylin, eosin, and DAB stains, and transformation from RGB to HED colour space using OD vectors and matrices provides a theoretical background for stain normalization [38]. Figure 2.4 sketches the absorbance–concentration relationship that motivates working in optical density for stain unmixing.

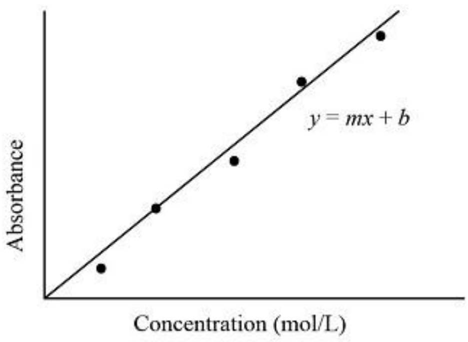

Figure 2.4. Beer–Lambert absorbance.

Other relevant stains include Immunohistochemistry (IHC), which serves as an additional staining method used to detect cellular behaviours that help laboratories further divide tumours into specific subcategories. IHC involves the use of specific markers such as ER, PR, HER2, and Ki67 for tumour categorization. IHC with anti-cytokeratin (CAM 5.2) is used for difficult cases to resolve diagnostic uncertainty. Immunohistochemistry is also used to estimate the number of cells positive for a particular antigen and the degree of positivity (staining intensity).

Phosphohistone-H3 (PHH3) staining can be used as a reference standard for mitosis detection when combined with H&E-stained slides. PHH3 is used to restain slides in combination with image registration to build reference standards for mitosis detection in H&E whole-slide images [38].

2.2.2 Sources of Stain Variability

Multiple sources contribute to stain variability in histopathology. Variability arises from differences in tissue preparation, staining protocols, dye compositions, and lab protocols. The appearance of stained slides varies among pathology laboratories and even within the same centre over time, indicating temporal variability in addition to spatial variability. Different histology protocols from different pathology labs contribute to this variability, as do variations in the digitization process due to different scanners [37].

Staining variation occurs across laboratories and over time within the same centre, representing a significant source of variability that affects algorithm performance. Poor staining quality can affect prediction performance, highlighting the practical impact of these variations. Variability due to lab protocols, reagents, and fixation methods affects both visual and algorithmic interpretation, leading to false positives and negatives in annotations [43].

Stain variability significantly impacts both human visual interpretation and algorithmic performance. Different staining characteristics can lead to batch effects and reduced accuracy in computational analysis. This variability causes algorithms to underperform on images from different laboratories, indicating a domain shift. Stain variability affects model performance and accuracy in computational pathology applications [31].

2.3 Digital Pathology Fundamentals

2.3.1 Definition of Whole-Slide Images

Digital pathology involves the digitization of tissue slides, enabling efficient storage, visualization, and advanced pathologic analysis [37]. Central to this process are WSIs, which are high-resolution digital representations of entire tissue sections, as shown in Figure 2.5. Produced by whole-slide scanners, WSIs allow pathologists to review slides on digital screens rather than through traditional microscopes. This transition to a digital workflow facilitates more accurate diagnostic processes, remote consultations, and easier access to specialist expertise [36].

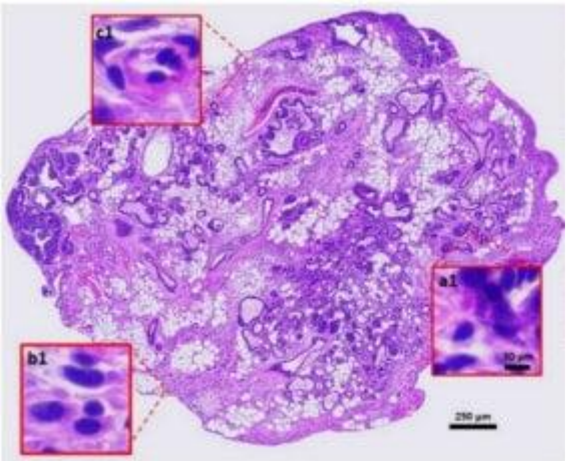

Figure 2.5. A whole slide image (WSI) of a breast tumour H&E [44].

2.3.2 Slide Digitization Workflow

The digitization workflow begins with tissue preparation, where sections are cut, stained, and scanned to produce high-resolution WSIs, as shown in Figure 2.6 [36]. This process, enabled by cost- and time-efficient scanners, aims to replace the optical microscope with a fully digital workflow encompassing tissue management, pathology orders, and reporting [41]. WSI resolution is determined by the specimen-level pixel size, which depends on the scanner and objective lens used (typically $20\times$ *or* $40\times$). At $40\times$ magnification, a common resolution of 0.25 μm/pixel is achieved for Aperio scanners, while Ventana systems reach 0.23 μm/pixel [38]

These images can reach dimensions of $80{,}000 \times 80{,}000$ pixels and storage sizes of approximately 20 GB [32]. The selection of appropriate magnification is critical for analytical tasks, as it dictates the available context for classification. While higher magnification often improves results through stronger signal strength, selecting a magnification that is too low can reduce analytical power by incorporating non-contextual pixels [34].

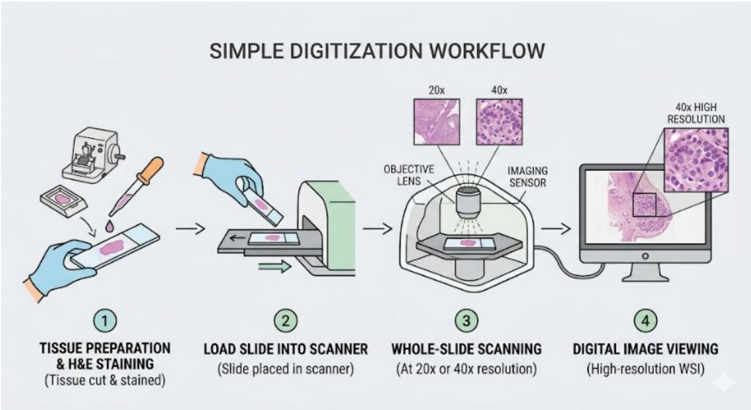

Figure 2.6. Simplified histopathology slide digitization workflow [36].

2.3.3 Computational Challenges of Whole-Slide Images

WSIs pose significant computational challenges due to their gigapixel scale, with resolutions up to $80{,}000 \times 80{,}000$ pixels, processing an entire slide at once is impractical. A standard CNN would require billions of parameters to handle such data, making direct application computationally infeasible and demanding vast resources for dense prediction. To address these requirements, innovative approaches like patch-based analysis are used, where large images are divided into smaller, manageable sections for detailed examination, as shown in Figure 2.7 [38].

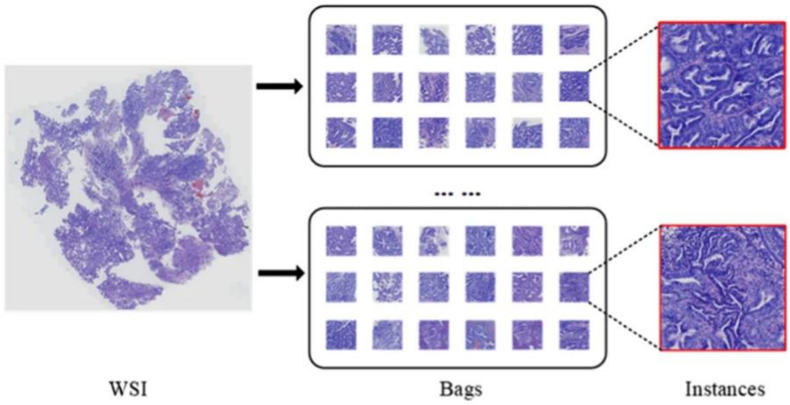

Figure 2.7. Illustration of the notions of WSI, bag, and instance (image patch) [45].

2.3.4 Patch-Based Analysis

Patch-based analysis is a fundamental paradigm in WSI processing where large images are divided into smaller, manageable sections for detailed examination. This approach addresses the computational infeasibility of processing entire gigapixel slides, which can contain billions of pixels and exceed the memory capacity of standard hardware [37]. In this workflow, convolutional neural networks (CNNs) are applied to small image tiles to estimate the probability of disease, such as separating tumour from normal tissue regions [1], [32], [4]. Generating diverse training patches is central to this method, often involving millions of tiles to ensure the model captures a wide range of tissue architectures. While this focuses the analysis on detailed local features, it can sometimes limit the broader architectural context [37]. Figure 2.8 depicts the patch-based paradigm: the slide is processed as a grid of small image tiles so that convolutional models can run within memory limits while still covering the whole section.

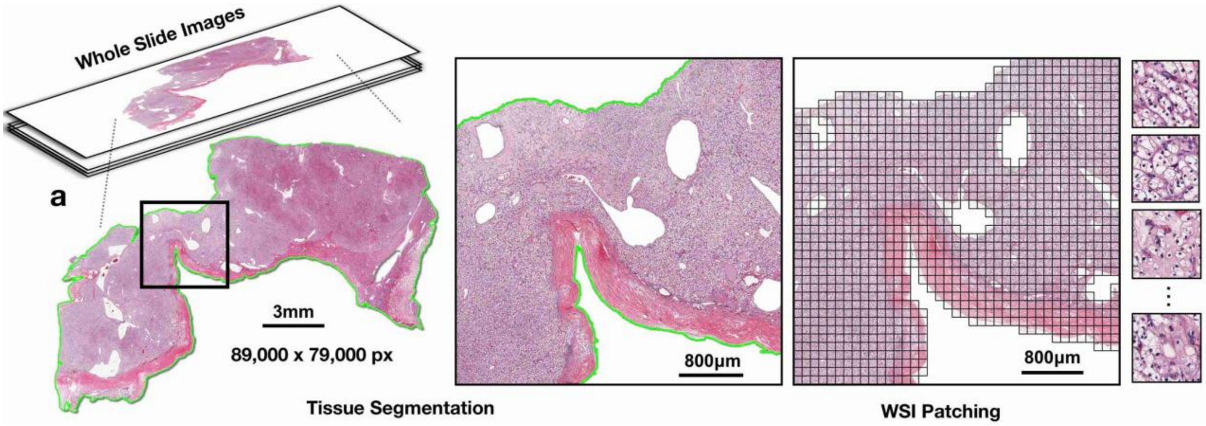

Figure 2.8. Whole-slide image split into patches [46].

The distinction between patch-level and slide-level prediction is essential for diagnostic accuracy. Patch-level analysis allows for granular tasks, such as identifying individual metastases or detecting mitotic cells within specific regions. In contrast, slide-level prediction involves aggregating information from all patches to determine if an entire slide is positive or negative for a condition [37]. Many studies utilize both paradigms—training models on image tiles to gain fine-grained information and then integrating these local detections into a final slide-level assessment [32]. While traditional CNNs are limited to these small regions, adaptive sampling methods can bridge the gap by focusing computational resources on regions of high uncertainty to better balance local detail with comprehensive whole-slide evaluation.

2.3.5 Scanner and Acquisition Variability

Variations in optics, sensors, and illumination across different scanner models, as shown in Figure 2.9, such as the Pannoramic 250 Flash II, NanoZoomer-XR, Aperio Scanscope CS, and Ventana iCoreo, significantly impact image consistency [37]. These hardware discrepancies affect the colour response and often introduce specific imaging artifacts [31]. To mitigate these effects, datasets often include images from multiple institutions. Beyond hardware, the appearance of stained slides varies among laboratories, leading to noticeable differences in intensity, colour range, and clarity [38]. The impact of scanner heterogeneity extends to variations in resolution and contrast, which can hinder both manual visual interpretation and automated computational analysis [43].

Figure 2.9. APERIO GT 450 Scanner [47].

2.4 Preprocessing Methods in Computational Pathology

Whole-slide image preprocessing addresses fundamental challenges arising from variability in tissue preparation, staining, and digitization. These variations introduce artifacts and colour inconsistencies that must be mitigated to ensure reliable downstream analysis [48].

2.4.1 Tissue Detection and Background Removal

Tissue localization is the initial step, isolating relevant regions from non-informative background areas. A standard approach, such as the CLAM toolbox, localizes tissue before extracting non-overlapping patches at specified magnifications [49]. Filtering background regions is crucial for computational efficiency, as background typically constitutes a substantial portion of the WSI [50].

2.4.2 Artifact Detection and Handling

Digital slides often contain artifacts introduced during preparation and digitization, such as blur, pen marks, tissue folds, and air bubbles. These can significantly degrade the performance of computational systems. Systematic detection and mitigation are required; for example, incorporating multiple prior knowledge of staining into the pipeline can reduce artifact-related failure rates to as low as 0.4% [48].

2.4.3 Patch-Level Quality Screening

At the patch level, simple quality gates remove tiles that are largely background, nearly uniform in intensity, or tissue-poor before stain normalization or training. Such rules reduce wasted model capacity on empty fields and make colour statistics more comparable across splits; they are not a substitute for pathologist review of difficult metastases.

2.4.4 Colour Normalization Approaches

Colour appearance varies across WSIs due to differences in specimen preparation, staining protocols, and scanner hardware. This heterogeneity impacts the performance of computer-aided diagnosis, necessitating normalization to ensure consistency [51], [52]. Stain normalization is essential for harmonizing histological images across different laboratories. These methods range from traditional mathematical deconvolution to advanced deep learning architectures, each offering unique technical advantages for addressing domain shift [53][54].

Traditional decomposition methods like the Macenko method utilize singular value decomposition (SVD) to estimate H&E stain vectors. By projecting data onto a plane derived from dominant SVD directions, it extracts stain absorbance coefficients and concentration maps to align images to a reference slide, effectively reducing intensity drift while preserving tissue structure [55], [56], [57]. Similarly, the Vahadane method decomposes images into sparse, non-negative stain density maps. By combining these maps with the colour basis of a target image, it adjusts colour without altering morphology; a style often preferred by pathologists in comparative studies. Faster variants of this method utilize patch subsampling to handle large whole-slide images effectively [58]. While Reinhard-style methods match basic colour statistics such as mean and variance, Macenko and Vahadane focus on estimating stain basis vectors through deconvolution [58], [59]. More complex probabilistic treatments, such as Bayesian blind deconvolution, employ Super-Gaussian and colour-vector priors to ensure higher fidelity to the original tissue structure [60]. Adaptive colour deconvolution (ACD) further enhances this by embedding multiple staining priors to jointly optimize separation and normalization [61].

These technical variations lead to measurable differences in performance across clinical benchmarks. For instance, the Macenko method has been shown to improve accuracy and F1-score in metastasis classification, while ACD increased cancer-classification AUC-ROC from 0.842 to 0.914 in a 500-WSI study [55], [62], [61]. Similarly, Bayesian normalization improved tissue fidelity and AUC on the Camelyon-16 and 17 datasets, while StainNet reported strong SSIM and PSNR values compared to alternatives [60]. These results underscore that while classical methods remain robust baselines, adaptive and learned approaches often provide superior generalization in multicentre diagnostic pipelines [61]. Figure 2.10 illustrates, on a single quality-screened PatchCamelyon tile, how the three classical colour-normalization families (Macenko, Reinhard, and Vahadane) reshape appearance when each is fit to the same fixed reference.

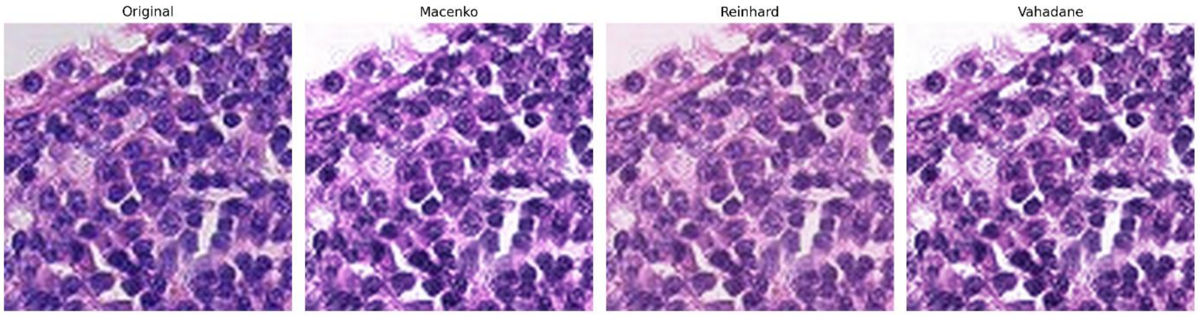

Figure 2.10. Same PCam patch (left) after Macenko, Reinhard, and Vahadane to a common reference.

The choice between single and multiple references significantly impacts model stability. A single target is standard for classical Macenko or Vahadane deployments but may under-represent the diversity of a large cohort. Conversely, randomly swapping references during training can destabilize optimisation. Multi-reference schemes, which estimate and combine stain matrices from several targets, offer a more robust solution. Averaging reference matrices has been reported as particularly effective when test staining diverges from the training set, and these modules can be integrated with minimal disruption to existing pipelines [57].

For downstream analysis, colour deconvolution is performed in optical density (OD) space rather than raw RGB space. This transformation is essential because light absorption by stains follows the Beer-Lambert law, allowing for the linear unmixing of individual stain contributions. By decomposing the image into its constituent components, individual stains can be analyzed and normalized separately before recombination. This structured approach to unmixing significantly improves the accuracy and consistency of subsequent classification tasks in computational pathology [51].

2.4.5 Data Standardization

Standardization of image resolution and size forms another critical preprocessing component. Preprocessing pipelines typically resize whole-slide images to standard resolutions, such as 0.5 μm per pixel at $20\times$ magnification [50]. Patches are extracted at consistent sizes, commonly 256×256 pixels at $20\times$ magnification [49]. These standardization steps ensure consistent input dimensions for downstream learning architectures while managing computational load.

2.5 Learning Architectures in Computational Pathology

Early computational pathology relied on manual feature engineering and traditional machine learning. A significant transition occurred with the introduction of 6-layer neural networks, which paved the way for modern Deep Convolutional Neural Networks (DCNNs) [63].

2.5.1 Convolutional Neural Networks for Histopathology

DCNNs shifted the paradigm by enabling automatic feature learning from raw data. However, while these architectures excel at local hierarchical representations, they often face generalization challenges when exposed to the technical variability of different scanner configurations [53]. Adapted training procedures, such as momentum-based stochastic gradient descent and batch normalization, are typically employed to refine these models for specific tasks like mitosis detection [63].

2.5.2 Vision Transformers and Self-Attention Mechanisms

Vision Transformers (ViTs) represent a shift toward self-attention mechanisms, allowing models to capture long-range dependencies and global context by tokenizing images into patches [50], [64]. Building on this, the Vision Mamba (Vim) architecture adapts state space models to capture these dependencies with greater efficiency at smaller scales, reporting an 8.21 increase in ROC AUC compared to standard ViTs [64]. For gigapixel-scale analysis, the GigaPath architecture utilizes LongNet methods to process tens of thousands of tiles per slide, employing both tile-level encoders for local features and slide-level encoders for global embeddings, Foundation Models, and Self-Supervised Pretraining [54]. Foundation models pretrained through self-supervised learning (SSL) currently represent the state-of-the-art for feature extraction. These models develop robust, generalizable representations from massive unlabelled datasets before being fine-tuned for clinical tasks [65]. UNI is pretrained on 100 million images from 100,000 diagnostic slides across 20 tissue types. It enables resolution-agnostic classification and generalizes across 108 cancer types [65]. Prov-GigaPath is developed using 1.3 billion image tiles from over 171,000 slides, achieving significant improvements in mutation prediction tasks [50]. Virchow Series, including Virchow 2 and 2G, scaled training to 3.1 million slides and 1.85 billion parameters, demonstrating that larger, domain-specific datasets consistently improve average diagnostic performance [66], [67]. These models substantially outperform generic versions by accelerating training convergence and enhancing the accuracy of downstream classification [49].

Frozen encoder training is a common deployment pattern for pathology foundation models that keeps the pretrained encoder fixed and trains only a small head on labelled patches from the target task. This limits how much the representation can adapt to a new scanner or hospital, but stabilizes training and isolates whether transfer failure arises from the backbone, the head, or preprocessing. Figure 2.11 diagrams the frozen-encoder fine-tuning layout: a pretrained pathology encoder supplies fixed features while only the classification head is updated on labelled patches. The present study follows this pattern for Virchow2 (frozen encoder, trainable linear head with dropout) while also reporting a fully trainable shallow CNN baseline on the same tensors.

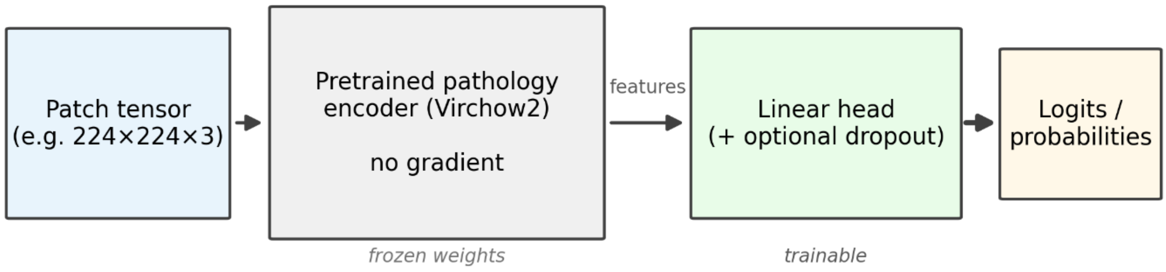

Figure 2.11. Frozen encoder with trainable linear head.

2.5.3 Weakly Supervised and Multiple Instance Learning for Whole-Slide Images

Because pixel-level annotations are costly and scarce, Multiple Instance Learning (MIL) is the dominant approach for whole-slide classification using only slide-level labels [49], [64]. In MIL, each slide is a bag of patch instances; features are extracted per patch and aggregated (for example with attention-based ABMIL or CLAM) into a slide-level decision [49]. The present project does not use MIL aggregation: every patch in PCam and CAMELYON17 (WILDS) already carries a patch-level metastasis label, so classifiers are trained and evaluated directly on tiles rather than on bags of unlabelled instances.

2.6 Domain Shift and Generalization

Domain shift occurs when differences in scanners, histology protocols, and staining cause algorithms to underperform on data from sources not seen during training [38]. These variations create batch effects and shifts in data distribution, which can prevent models from identifying rare events if they have only been exposed to specific tissue architectures or stain profiles [32], [43]. The distribution of input appearance (tissue colour and texture) may differ between training and testing environments, even when diagnostic labels remain defined in the same way [53], [63]. Figure 2.12 illustrates domain shift as a mismatch between the distribution of training inputs and that of test or deployment inputs. Stain-related causes are introduced in Sources of Stain Variability; scanner and acquisition effects are introduced under Digital Pathology Fundamentals.

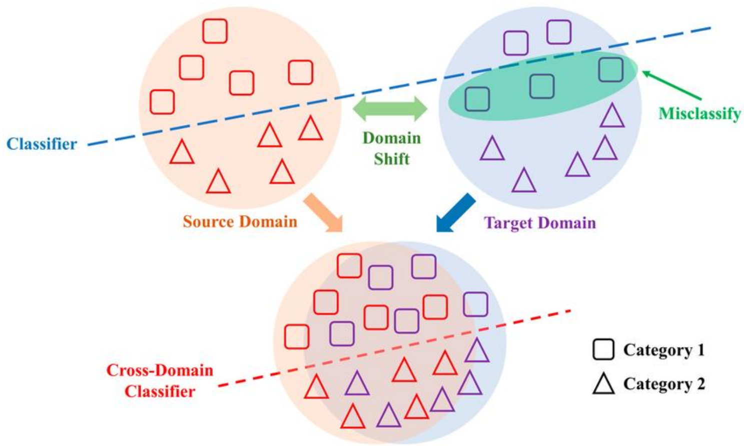

Figure 2.12. Domain shift: train vs test distributions [68].

In-domain performance refers to evaluation on data drawn from the same distribution as the training set, whereas out-of-domain (external) performance refers to evaluation on unseen hospitals, scanners, or stain profiles. Without explicit mitigation, machine learning algorithms tend to overfit to specific centres or scanners and fail on external sources [49], [53], [63].

Mirrored external validation (train on A, test on B, and the reverse) shows whether transfer loss is symmetric and how probability behaviour differs by direction. A model trained on benchmark A and evaluated on benchmark B may not match the reverse when trained on B and tested on A, because stain profiles, class balance, and hospital composition differ. In this thesis, the forward leg from PCam to CAMELYON17 carries the principal external question; the reverse leg is reported to document asymmetry and calibration effects, not as a second symmetric requirement for the forward claim. Reporting both directions and the drop from in-domain to external test on the same metric set still sharpens interpretation relative to a single convenient one-way report.

Domain adaptation (e.g., DANN) assumes access to unlabelled target data during training. DANN uses a gradient reversal layer to maximize classification accuracy while preventing the feature extractor from encoding domain-specific information [53]. Domain generalization (e.g., the WSI-P2P framework) trains models to perform on unseen domains without target data at training time [53], [63], [34]. Stain normalization and quality-aware preprocessing (Preprocessing Methods) are complementary ways to narrow appearance gaps before classification.

Generalization in computational pathology is driven by dataset scale and diversity, training-time colour augmentation, and pairing preprocessing with appropriate encoders. Large pathology foundation models pretrain on broad slide collections and often transfer more effectively than models trained on narrow corpora alone, but they do not remove the need for careful external testing or calibration when the deployment site changes [19], [22], [33], [35].

Clinical deployment robustness emerges from the strategic layering of multiple techniques rather than relying on a single solution. Because no universally superior technique exists for all histological contexts, a multi-faceted framework is required to achieve the stability demanded by real-world diagnostic environments [53]. Beyond the generalizable base representations provided by large-scale foundation models, online adaptive feature extraction allows systems to fine-tune to specific deployment contexts without catastrophic forgetting. This mechanism addresses multi-centreed variability while maintaining a robust performance baseline [49], [34].

Furthermore, integrating advanced preprocessing modules, such as adaptive colour deconvolution, can reduce artifact-related failure rates to as low as 0.4% [52].When combined with structure-preserving normalization methods that align with pathologist preferences, these modules ensure that digital slides are both computationally reliable and visually consistent [58]. The final optimisation layer involves targeted domain adaptation when specific deployment sites are known, creating a comprehensive pipeline that integrates feature extraction, efficient aggregation, and robust preprocessing for scalable clinical use [16], [27], [34].

To combat these shifts, several technical strategies are employed to improve model robustness and ensure reliable performance. Training models on diverse, multicentre datasets from various sites and platforms ensures better generalization across different acquisition conditions [35]. Furthermore, data augmentation techniques like StyleGAN and heavy colour augmentation, specifically modifying H&E colour channels, simulate realistic stain variations to prepare models for real-world inconsistencies [31], [38]. Finally, the implementation of generalizable stain normalization and standardization allows for the automatic processing of WSIs without the need for manual institutional calibration to specific stain profiles.

2.7 Uncertainty and Calibration

Uncertainty in medical AI models comprises two fundamental types with distinct implications. Epistemic uncertainty represents model-inherent uncertainty that is reducible through increased training data, improved architectures, or refined training procedures. It directly relates to the model's prediction confidence, indicating the algorithm's certainty in its outputs [35]. This type of uncertainty can be addressed through Bayesian deep learning approaches that model distributions over model weights [36]. Aleatoric uncertainty, conversely, arises from inherent noise in input data and remains irreducible regardless of model improvements [35], [36]. The distinction is critical for clinical applications: epistemic uncertainty indicates where a model requires better training, while aleatoric uncertainty highlights cases with inherently ambiguous or low-quality data [35]. This enables clinicians to differentiate between algorithmic limitations and data-driven uncertainty.

2.7.1 Bayesian Formulations and Probabilistic Models

Bayesian frameworks provide the mathematical ground for uncertainty quantification by modeling distributions over model weights [37], [36]. The predictive entropy used to quantify this uncertainty is defined as:

$$H(y|x,D) \approx -\sum_{c=1}^{C} \overline{p}_c \log(\overline{p}_c) \tag{1}$$

where H is the predictive entropy (measure of uncertainty), C is the total number of classes $\bar{p}_c$ The average probability for class c across T.

The Monte Carlo (MC) dropout framework is a standard approximate Bayesian inference method used for epistemic uncertainty assessment. This technique involves maintaining dropout during both training and inference, using specified rates to sample different network configurations through stochastic dropout masks [35], [36]. By performing multiple forward passes for a single input, the model effectively samples from the prediction distribution, performing Bayesian model averaging without the need for complex variational inference [35].

While MC dropout is a computationally efficient way to estimate epistemic uncertainty, its calibration performance in digital pathology has been found to be inferior to Deep Ensembles [38].

Monte Carlo samples are calculated as:

$$\frac{1}{T}\sum_{t=1}^{T} \hat{p}_{c,t}.T \tag{2}$$

where T is the total number of Monte Carlo samples, t is the iteration index. $\hat{p}$ is the predicted probability and c is the class index which refers to a specific category or class in a classification task. This approach enables approximate Bayesian inference without requiring explicit prior distributions or complex variational inference procedure [35].

2.7.2 Probability Quality Metrics

Probability quality metrics evaluate the reliability of the predicted probabilities themselves rather than just the final binary label. The Brier Score (BS) measures the mean squared difference between predicted probabilities and actual outcomes [36].

The Brier Score is defined as

$$BS = \frac{1}{N}\sum_{i=1}^{N} (f_i - y_i)^2 \tag{3}$$

Where N is the total number of samples, f_i is the predicted probability for sample i and y_i is the actual binary label (0 or 1) for sample i.

Log loss (logarithmic loss) penalizes confident incorrect predictions more heavily than less confident errors, though it is not explicitly detailed in the included abstracts.

$$\text{LogLoss} = -\frac{1}{N}\sum_{i=1}^{N} [y_i \log(p_i) + (1 - y_i)\log(1 - p_i)] \tag{4}$$

Where p_i is the predicted probability of the sample belonging to the positive class.

2.7.3 Calibration Methods

Calibration refers to the alignment between a model's predicted confidence and its actual correctness. Miscalibration is the deviation between predictive probability and classification accuracy [39].

The Expected Calibration Error (ECE) serves as the primary metric for assessing calibration using the mathematical formulation:

$$ECE = \sum_{m=1}^{M} \frac{|B_m|}{N} |acc(B_m) - conf(B_m)| \tag{5}$$

where M is the total number of bins used to group confidence scores. B_m is the set of indices of samples whose prediction confidence falls into the m^{th} bin. $|B_m|$ is the number of samples in bin m. N is the total number of observations in the dataset. $acc(B_m)$ is the average accuracy of predictions in bin m. $conf(B_m)$ is the average confidence (predicted probability) in bin m [38].

Additional metrics include Overconfidence Error (OE), which quantifies confident but incorrect predictions, and Maximum Calibration Error (MCE), which identifies the largest deviation across all bins. The Brier Score (BS) is also used to evaluate the overall accuracy of probabilistic predictions. While these metrics collectively assess different aspects of calibration, research indicates that inconsistencies can arise when using different measures across the same model [36].

Temperature Scaling is the most widely studied post-hoc calibration method. It adjusts output probabilities by learning a single scalar parameter, T on a validation set to scale the logit vectors before the softmax layer [39], [38]. The calibrated probability $\hat{q}_i$ is calculated as:

$$\hat{q}_i = \max_k \sigma(z_i/T)^{(k)} \tag{6}$$

where z_i is the logit vector (raw output) for sample i. T is the scalar temperature parameter (optimized on a validation set). σ is the softmax function. k is the class index. This method has shown strong results in dermatopathology and histopathology, especially when applied to networks trained with focal loss [39]. An advanced variant, Dual Temperature Scaling (DTS), employs two temperature parameters to refine the posterior distribution, providing more flexible adjustments and improving model explainability [37].

Beyond post-hoc adjustments, training-time strategies modify the objective function to encourage well-calibrated predictions. Confidence-Uncertainty Boundary Loss (CUB-Loss) imposes penalties on high-certainty errors and low-certainty correct predictions to enforce alignment between correctness and uncertainty [37]. Confidence Weighting: Weights sample losses to explicitly penalize confident incorrect predictions. This approach has been shown to reduce ECE by up to 22% in cardiovascular diagnostic tasks while slightly improving overall accuracy [36]. And robust Calibration Regularization (RCR): An auxiliary loss function that regularizes density estimation using a parametric prior model. This specifically addresses the high bias and variance often found in calibration metrics when working with small medical imaging datasets [40].

Deep Ensembles involve training multiple networks individually and averaging their predictions during inference. This approach demonstrates superior robustness toward domain shifts and label noise compared to baseline models [38], [41]. The ensemble approach enables epistemic uncertainty estimation through variance in predictions across models [36]. Mixup is a training-time data augmentation technique that improves calibration by creating synthetic training examples through linear interpolation. It combines random pairs of images and their labels, forcing the model to behave linearly in-between training samples [38]

The synthetic sample ($\bar{x}$,$\tilde{y}$) is generated as:

$$\bar{x} = \lambda x_i + (1 - \lambda)x_j \tag{7}$$

$$\tilde{y} = \lambda y_i + (1 - \lambda)y_j \tag{8}$$

where x_i , x_j are raw input vectors (images). y_i, y_j are one-hot encoded labels. λ is a mixing proportion sampled from a Beta distribution, $\lambda \in [0, 1]$.

2.8 Evaluation Metrics

Classification metrics quantify model performance for categorical prediction tasks. Standard metrics include accuracy, precision, recall, and F1 score; are derived from the confusion matrix components: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{9}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{10}$$

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN} \tag{11}$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{12}$$

where, TP is True Positives (Correctly predicted positive cases), TN is True Negatives (Correctly predicted negative cases), FP is False Positives (Type I Error; negative cases predicted as positive) and FN is False Negatives (Type II Error; positive cases predicted as negative).

The choice of classification metric depends on clinical context. For imbalanced datasets common in medical imaging, accuracy alone may be misleading. Precision quantifies the proportion of positive predictions that are correct, while recall measures the proportion of actual positives correctly identified. The F1 score harmonically balances precision and recall, though it is not explicitly reported in most included studies [36], [38]. Ranking metrics assess a model's ability to order predictions by confidence or probability. The two primary measures are the Area Under the Receiver Operating Characteristic curve (AUROC) and the Area Under the Precision-Recall curve (AUPR).

$$AUROC = \int_0^1 TPR(FPR)dFPR \tag{13}$$

where $TPR = \frac{TP}{TP+FN}$ is (Sensitivity/Recall) and $FPR = \frac{FP}{FP+TN}$ is (1 - Specificity). Figure 2.13 recalls the ROC construction used later when reporting discrimination.

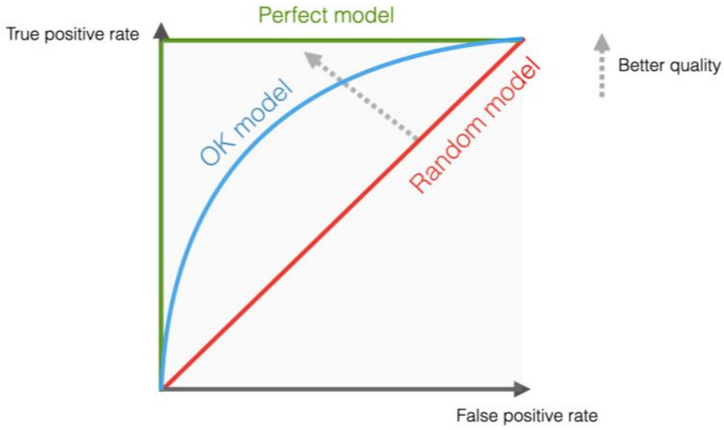

Figure 2.13. ROC: TPR vs FPR [77].

Area Under the Precision-Recall curve (AUPR or PR-AUC) represents the two primary ranking metrics.

$$AUPRC = \int_0^1 P(R)\ dR \tag{14}$$

where $P = \frac{TP}{TP+FP}$ is (Precision) and $R = \frac{TP}{TP+FN}$ is (Recall).

The Area Under the Precision-Recall Curve is utilized (AUC-PR) as a performance measure to evaluate uncertainty map quality [35]. The AUC-PR assesses how well uncertainty maps predict misclassification, independent of calibration. Both AUROC and AUPR are employed for misclassification and out-of-distribution detection [38], enabling comparison of methods' discriminative capabilities. The distinction between ROC-AUC and PR-AUC is clinically significant. ROC-AUC considers both true positive and false positive rates across thresholds, while PR-AUC focuses on precision and recall, making it more informative for imbalanced datasets where positive cases are rare. Ranking metrics such as ROC-AUC and PR-AUC are unchanged by strictly monotone transforms of logits (including temperature scaling), but threshold-based metrics (accuracy, recall, and F1 at a fixed cut-off such as 0.5) depend on the calibrated probability scale. A model can therefore rank well on external data while still failing clinically relevant sensitivity if probabilities are miscalibrated; both views are reported in the results chapter.

In medical contexts with heavy class imbalance, standard accuracy is often biased toward the majority class. Balanced Accuracy mitigates this by averaging the per-class errors, providing an equal weight to both Sensitivity (TPR) and Specificity (TNR) [42], [43]. Specificity is defined as:

$$Specificity = \frac{TN}{TN + FP} \tag{15}$$

It is a critical metric when the clinical cost of False Positives is high. In a 95% majority-class scenario, a model that simply predicts the majority class would achieve high accuracy but only 50% balanced accuracy, exposing its failure to identify the minority class [43].

The **MCC** is a correlation-style score that summarizes the full confusion matrix into a single value, ranging from -1 to +1. It is particularly suited for imbalanced binary problems because it requires high performance in all four quadrants (TP, TN, FP, FN) to achieve a high score [42], [43], [44].

$$MCC = \frac{(TP \times TN - FP \times FN)}{\sqrt{[(TP + FP)(TP + FN)(TN + FP)(TN + FN)]}} \tag{16}$$

A high MCC ensures that the model is not merely relying on the majority class route. Its acceptance in high-stakes regulatory settings underscores its reliability for quality-control style analyses [44].

For heavy class skew, precision–recall summaries are often preferred because they are less dominated by true negatives than ROC-based summaries [45]. ROC AUC and equal error rate are listed among standard discrimination tools. The MCC–F1 curve combines MCC and F1 to separate models under imbalance more clearly than either alone [46].

2.9 Calibration Metrics

While the Expected Calibration Error is the most widely adopted metric, it can suffer from high bias and variance in small medical datasets. To address this, the Robust Expected Calibration Error (RECE) was proposed to regularize density estimation using a parametric prior, providing more stable estimates for clinical imaging [40]. Other specialized metrics include Maximum Calibration Error (MCE) which identifies the largest deviation across all confidence bins, representing the "worst-case" scenario for clinical safety [36].

$$MCE = \max_{m \in \{1,\ldots,M\}} |acc(B_m) - conf(B_m)| \tag{17}$$

where max is the operator selecting the largest difference across all M bins. The Overconfidence Error specifically penalizes bins where confidence exceeds accuracy, targeting the most dangerous type of miscalibration

$$OE = \sum_{m=1}^{M} \frac{|B_m|}{N} [conf(B_m) \times max(conf(B_m) - acc(B_m), 0)] \tag{18}$$

where $max(\ldots, 0)$ Ensures only bins where confidence exceeds accuracy (overconfidence) are penalized.

Reliability diagrams provide visual representations of calibration by plotting predicted confidence against observed accuracy. Reliability diagrams are used to visualize calibration performance, specifically employing Adaptive ECE (AECE) for the CAD diagnosis task [36], [47]. These diagrams enable intuitive assessment of whether models are over-confident, under-confident, or well-calibrated across different confidence ranges. Figure 2.14 shows the reliability-diagram view of calibration: where predicted confidence tracks observed accuracy and where the model tends to be over- or under-confident.

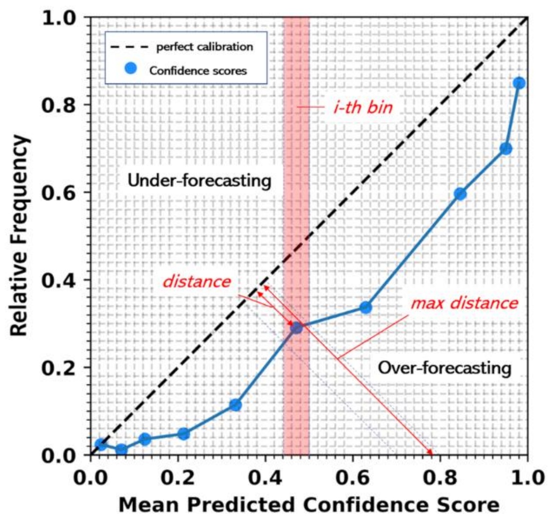

Figure 2.14. Reliability diagram: confidence vs. accuracy [84].

2.10 Operational Calibration and Clinical Decision Support

The choice of an operating threshold involves balancing competing clinical objectives based on the model's intended role. For screening applications, high sensitivity is paramount to ensure comprehensive disease detection, even if it results in higher false positive rates. Conversely, for confirmatory tests, high specificity is prioritized to avoid the physical and financial costs of unnecessary interventions. Because different calibration measures can identify different "optimal" models, selection criteria must be explicitly aligned with these clinical requirements [36].

Confidence quality is paramount in clinical decision support systems, as miscalibrated models can lead to erroneous clinical outcomes. Ideally, AI classification models should maximize confidence in correct predictions while avoiding "confident wrong" predictions [36]. Models that achieve this are considered well-calibrated, meaning their uncertainty estimates correlate directly with prediction correctness. This allows clinicians to identify unreliable outputs for human review, which is a fundamental requirement for the safe deployment of AI in critical tasks like medical imaging [37], [40].

The clinical implications of this calibration extend to automation and risk management. Selective decision-making enables a system to defer uncertain cases to experts, thereby protecting the diagnostic process from overconfident algorithmic mistakes that might otherwise erode user trust [38]. Furthermore, confidence quality must account for label disagreement between human experts. Calibrating models to reflect scenarios where the ground truth itself is uncertain addresses distributional uncertainty and inter-rater disagreement, ensuring the model's output mirrors the underlying clinical complexity [48].

Beyond aggregate metrics, case-based review links predictions to tissue appearance. Typical workflows export misclassified or high-uncertainty patches with labels, probabilities, and preprocessed images, then apply a fixed checklist (for example tissue amount, stain atypia, small-focus pattern, or patch-context limits). Prevalence in reviewed buckets supports interpretation of failure modes but does not replace quantitative transfer analysis on the full test split.

Chapter 3 Literature Review

This chapter traces how research on automated lymph-node metastasis detection moved from early image analysis through digital whole-slide imaging and patch-based benchmarks to modern convolutional and foundation-model approaches. It reviews clinical and dataset context for sentinel-node assessment, the Camelyon and PatchCamelyon corpora and related patch-level work, stain variability together with normalization and augmentation strategies, and the emergence of large-scale self-supervised pathology encoders and their comparative evaluation.

3.1 Lymph-Node Metastasis, Digital Pathology, and Patch Benchmarks

Breast cancer staging depends critically on histopathological assessment of sentinel lymph node biopsies, where pathologists examine H&E stained tissue sections for metastatic deposits. This process underpins the TNM classification system and directly informs treatment decisions [8]. Figure 3.1 summarizes, in schematic form, how sentinel lymph node assessment fits into the staging pathway that motivates automated assistance in digital pathology.

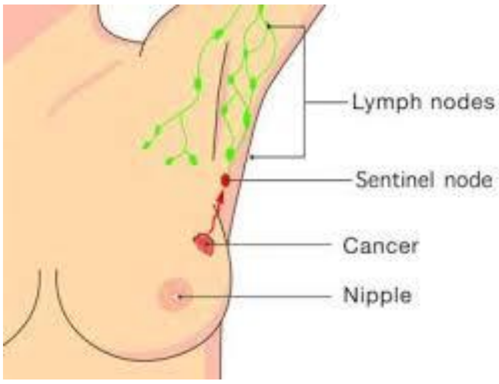

Figure 3.1. Sentinel lymph node and staging context (schematic) [86].

The first automated approaches in pathology did not begin with deep learning; they began with morphometry and computer-assisted image analysis. Early work in the 1990s argued that image analysis could provide an objective basis for histopathologic decision-making, and by the late 1990s and early 2000s, reviews of automated histometry and quantitative histology described systems that segmented tissue, measured nuclear and glandular features, and used those measurements to support diagnosis [87], [88].

By the time digital whole-slide images became practical, the field had already moved toward computer-aided diagnosis as a formal research area. Reviews from the 2000s and 2010s describe how digital scanners, image analysis, and machine learning turned histopathology into a true computational problem, setting up the later transition to slide-level deep learning and benchmark datasets [89], [90]

The transition from traditional microscopy to digital pathology introduced a fundamental computational challenge. A single whole-slide image (WSI) digitized at high resolution (e.g., 240 nm/pixel) is a multi-gigapixel image that cannot be loaded into standard GPU memory in its entirety [91]. For context, the CAMELYON dataset alone, comprising 1,399 WSIs from breast cancer patients, consumes roughly 2.95 terabytes of storage [21]. Because processing these images whole is computationally infeasible, researchers developed patch-based paradigms: WSIs are tiled into small patches (typically 256×256 pixels, with PCam using 96×96), classified individually, and aggregated into slide-level predictions. Figure 3.2 illustrates the standard multi-resolution pyramid used in digital pathology, which makes it practical to store, view, and algorithmically process slides that are far too large to handle as a single full-resolution array.

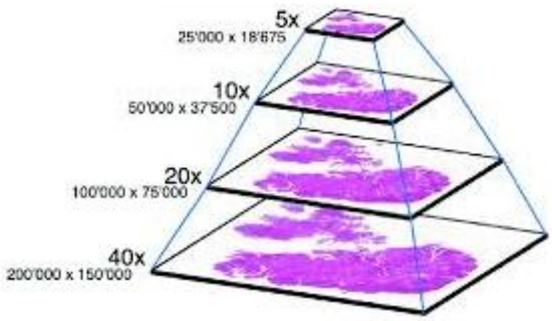

Figure 3.2. WSI multi-resolution pyramid [92].

The Camelyon Grand Challenges were established to catalyze automated solutions. CAMELYON16 demonstrated that the best deep learning algorithms could achieve an AUC of 0.994, outperforming pathologists under simulated time constraints [8]. A landmark study demonstrated through the CAMELYON16 challenge that the top-performing deep learning algorithm achieved an AUC of 0.994 for slide-level metastasis classification, significantly outperforming a panel of eleven pathologists (mean AUC 0.810). CAMELYON17 extended this to patient-level pN-staging across five centres, highlighting challenges in detecting Isolated Tumour Cells (ITCs). To facilitate patch-level research, PatchCamelyon (PCam) was introduced, a dataset of 327,680 labelled tiles extracted from CAMELYON16 slides. PCam has since become a standard benchmark for histopathology feature extraction [93]. CAMELYON17 is also distributed in the WILDS benchmark collection as patch-level data from multiple hospitals. Recently, these datasets were systematically reprocessed with expert pixel annotations to re-evaluate modern foundation models and Multiple Instance Learning (MIL) methods [21].

3.2 Stain Handling and Colour Space Augmentation

A fundamental challenge in computational pathology, not only in breast cancer metastasis detection but across organ types and cancer subtypes, is that H&E stained slides exhibit substantial colour variation across institutions, arising from differences in staining protocols, reagent brands, tissue preparation, section thickness, and scanner hardware. This variation degrades model performance on unseen institutions, a problem known as domain shift. Figure 3.3 gives a concrete impression of how much H&E appearance can vary across sites and instruments, which underpins the stain-handling literature summarized below.

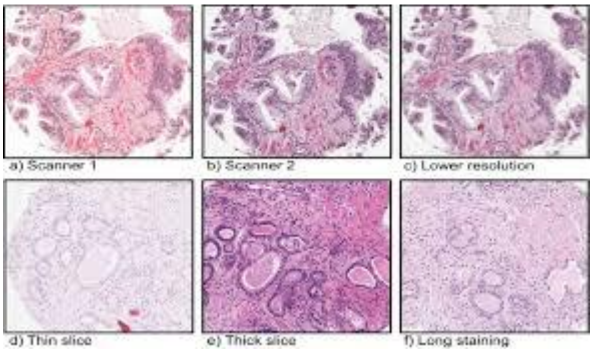

Figure 3.3. H&E appearance variability across settings [94].

Several classical stain normalization methods, such as those proposed by Macenko, decompose H&E images into stain vectors to map them to a reference template [95],[96]. The Macenko method works by decomposing H&E images into stain vectors via singular value decomposition in optical density space, then mapping to a reference template [95]. This method remains widely used for its simplicity. The Vahadane method extended this using sparse non-negative matrix factorization, which better preserves tissue morphology. Comparative evaluations show Vahadane achieves higher structural fidelity (SSIM), while Macenko provides more consistent chromatic distributions [96]. At the whole-slide level, WSICS is a fully automated algorithm that aligns chromatic and density distributions per stain component to a template WSI, achieving the lowest colour variation across 125 WSIs from 5 laboratories [97]. A comprehensive review in Information Fusion, grouping methods into four categories, showed that structure-preserving unified transformation-based methods consistently outperform alternatives [98].

Stain augmentation, normalization, and a neural network-based normalization were compared on heterogeneous data from 9 laboratories and 4 organs. Their key finding was that extensive stain augmentation in H&E colour space was more effective than normalization alone for CNN generalization [99]. Augmentation directly in H&E colour space (decomposing via colour deconvolution, modifying H and E channels stochastically, then projecting back to RGB) significantly improved robustness to unseen stain variations [100].

Six configurations combining Macenko and Vahadane normalization with colour augmentation on PCam were compared. Macenko combined with colour augmentation improved accuracy by 1.59% and F1 by 2.50% over baseline, modest gains reflecting PCam's relatively homogeneous origin, but consistent [101]. CAGAN is a colour adaptive generative network combining supervised and unsupervised learning, reporting 5–10% improvement on CAMELYON16, CAMELYON17, and other datasets over baselines without normalization [102]. Bayesian blind colour deconvolution improved AUC from 0.9491 to 0.9656 on Camelyon-16 and from 0.9279 to 0.9541 on Camelyon-17 [60]. Colour normalization improved retrieval accuracy to 97% on CAMELYON17, a 14% increase [103]. Aggregating predictions from the two best normalization methods with augmentation improved external-set AUC from 0.860 to 0.892, confirming preprocessing benefits are most pronounced when generalizing across institutions [104].

HistAuGAN is a structure-preserving multi-domain stain augmentation using disentangled representations. When evaluated on Camelyon17, it outperformed conventional colour augmentation and mitigated batch effects [105]. Learnable preprocessing operations trainable end-to-end with classification networks were also proposed [106]. Domain-specific modifications to RandAugment for H&E achieved AUC of 0.964 on Camelyon17 versus 0.958 for manual augmentation [107], [108].

AIDA (Adversarial Fourier-based Domain Adaptation) was proposed, which significantly outperformed both colour normalization and standard adversarial approaches across four cancer types from multiple centres, suggesting stain normalization alone is insufficient for full cross-centre generalization [109]. Additionally, combining stain normalization with segmentation-driven patch extraction was shown to improve accuracy by 17–18% for prostate and breast cancer detection [110].

Most normalization methods require selecting a reference image, introducing subjectivity. GAN-based methods risk mode collapse or information loss. The interaction between normalization and modern foundation models remains underexplored, whether models pretrained on millions of diverse slides already encode stain-invariant features, rendering explicit normalization redundant, is an open question.

3.3 From Handcrafted Features to Convolutional Patch Classifiers

During this Pre-deep learning era, computational pathology relied on traditional machine learning, where researchers manually extracted morphological and texture features, such as Local Binary Patterns (LBP), Scale-Invariant Feature Transform (SIFT), and Grey-Level Co-occurrence Matrices (GLCM), and classified them using Support Vector Machines (SVMs), Random Forests, and other algorithms [111].

While these methods demonstrated that automated detection was possible across various cancer types (breast, colourectal, prostate, gastric), they were heavily dependent on human intuition for feature selection and struggled to capture the immense heterogeneity of tumour microenvironments [112].

These standardized benchmarks catalyzed a diverse range of algorithmic strategies. With the emergence of CNNs, models could automatically learn hierarchical features directly from raw pixel data, eliminating the need for handcrafted feature engineering. On PCam, a modified VGG16 achieved an AUC of 0.90 [113], while hybrid models ensembling DenseNet169 and ResNet50 demonstrated that dense connectivity could effectively alleviate the vanishing gradient problem in deep pathology networks. Other specialized approaches, such as semi-supervised learning with pseudo-labels and supervised contrastive learning (e.g., HistopathAI), further improved performance and highlighted the generalizability challenges across multiple cancer types [114]. Despite these advances, researchers identified significant domain-specific limitations in standard CNNs. First, histopathology images exhibit inherent rotational and reflectional symmetries; tissue appears equally valid at any orientation, yet standard CNNs often waste parameter capacity learning these variations redundantly. To address this, P4M-DenseNet, a rotation-equivariant CNN, was introduced alongside the PCam benchmark to explicitly encode these symmetries, resulting in more stable predictions [115]. This concept was further extended with Dense Steerable Filter CNNs, achieving state-of-the-art results with significantly fewer parameters [116].

Furthermore, isolated patch analysis often lacks the global spatial context necessary for accurate diagnosis. The MMSEN (Multi-scale Multi-head Self-attention Ensemble Network) addressed this by integrating self-attention mechanisms to guide the model toward critical tumour-related features, reaching a ROC-AUC of 99.01% on PCam [117], [118].

A pivotal shift occurred with the adoption of ImageNet-pretrained networks. Despite the large visual domain gap between natural photographs and microscopy, fine-tuning architectures like Inception-V3 and ResNet50 proved highly effective, with ResNet50 achieving 97.50% accuracy on breast histology subtypes [119]. Comparative studies confirmed that fine-tuned networks substantially outperform those trained from scratch [120]. Fine-tuned architectures like VGG16 yielded 92.60% accuracy, significantly outperforming models trained from scratch [29], while pre-training on domain-specific histopathology data showed even greater gains [30]. However, the visual dissimilarity between natural images and histopathology remained a persistent concern. Comparisons between domain-specific and ImageNet weights revealed that pathology-specific pretraining yields higher performance and better feature reuse, foreshadowing the industry-wide shift toward large-scale pathology foundation models [121].

3.4 Vision Foundation Models and Representation Capacity

The scarcity of labelled histopathology data, where annotation requires expert pathologists, motivated self-supervised learning (SSL) approaches that could leverage the vast quantities of unlabelled WSIs available across institutions. This "annotation bottleneck" was a persistent limitation of fully supervised approaches. The power of SSL for clinical applications was demonstrated by utilizing the Bootstrap Your Own Latent (BYOL) framework to accurately identify eyelid malignant melanoma using only a fraction of the labelled data previously required, confirming that self-supervised features transfer effectively to rare pathology tasks [122]. When contrastive self-supervised learning across 60 histopathology datasets without labels is applied, it was found that histopathology-pretrained networks outperformed ImageNet-pretrained ones by up to 7.5% in accuracy and 8.9% in F1 [123].

CTransPath is a hybrid CNN-Transformer backbone pretrained using semantically-relevant contrastive learning (SRCL). It achieved state-of-the-art on nine public datasets spanning five task types, outperforming both ImageNet-supervised and self-supervised pretraining [124]. UNI is pretrained on over 100 million images from 100,000+ diagnostic WSIs across 20 tissue types. Evaluated on 34 tasks, UNI outperformed prior state-of-the-art and demonstrated resolution-agnostic classification and few-shot slide classification [125]. CONCH is a visual-language foundation model trained on over 1.17 million image-caption pairs. CONCH achieved state-of-the-art on 14 diverse benchmarks spanning classification, segmentation, captioning, and retrieval [126]. Prov-GigaPath is a whole-slide pathology foundation model pretrained on 1.3 billion 256×256 pathology image tiles from 171,189 WSIs covering 31 tissue types. Using a novel vision transformer architecture adapted from LongNet for ultra-large-context modeling, GigaPath achieved state-of-the-art on 25 out of 26 tasks in their benchmark [127]. Virchow is trained on 1.5 million WSIs from 100,000 patients. It achieved 0.95 specimen-level AUC for pan-cancer detection across nine common and seven rare cancers, and showed that with limited labelled data, Virchow-based detectors matched tissue-specific clinical-grade models [128]. Virchow2 (632M parameters), Virchow2G (1.9B), and a distilled 22M variant, trained on 3.1 million WSIs. They achieved state-of-the-art on 12 tile-level tasks and showed that combining domain-specific methods, data scale, and model scale yields the best performance [129].

The rapid proliferation of foundation models created urgent need for independent evaluation. When 19 models across 13 cohorts (6,818 patients, 9,528 slides) were benchmarked, CONCH yielded the highest performance, Virchow2 close second. An ensemble of CONCH and Virchow2 outperformed individual models in 55% of tasks [130]. Furthermore, 31 models across 41 tasks were benchmarked, finding Virchow2 highest overall and that model/data size did not consistently correlate with performance [130]. PathBench (presented across 15,888 WSIs from 10 hospitals) found Virchow2 and H-Optimus-1 the most effective [131].

3.5 Downstream Slide-Level Aggregation

While patch-level classification is the building block, clinical decisions require slide-level or patient-level predictions. MIL treats each slide as a "bag" of patch instances. Figure 3.4 depicts the weakly supervised slide-level setting in which only slide labels are available while models consume patch tiles. A proposed MIL-CNN framework achieved up to 89.52% accuracy for breast tumours [132]. A Prototypical MIL using unsupervised clustering achieved AUC of 88.2% on Camelyon16 and outperforming existing MIL methods for micro-metastases [133]. Modern pipelines now use foundation model features (from UNI, GigaPath, CTransPath, etc.) fed into advanced MIL aggregators such as ABMIL, CLAM, or TransMIL to generate slide-level predictions [134]. The present project evaluates patch-level metastasis classification on PCam and CAMELYON17 (WILDS) directly rather than slide-level MIL aggregation.

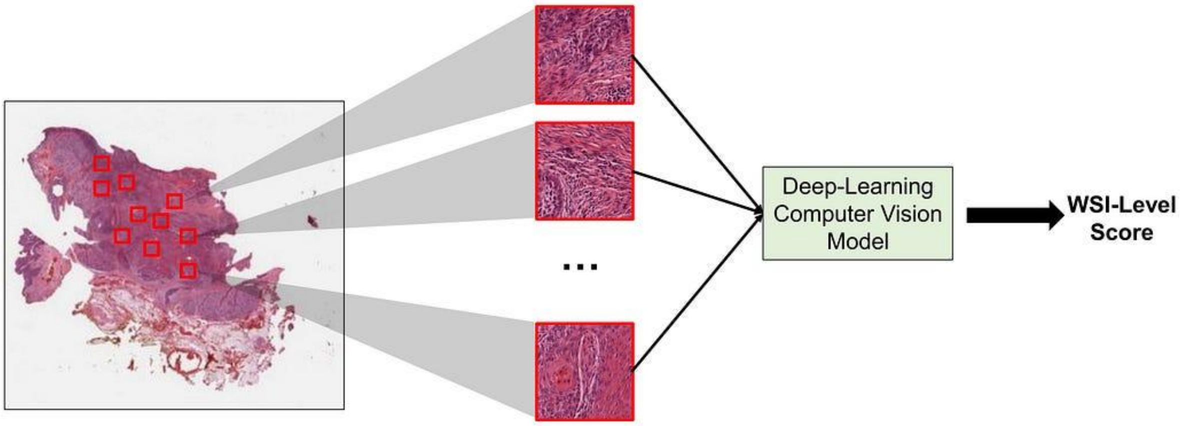

Figure 3.4. MIL: slide as a bag of patches [135].

3.6 Research Gaps

Despite considerable progress, several gaps remain. The interaction between preprocessing pipelines and foundation model features is largely untested, whether explicit normalization adds benefit when using models pretrained on millions of diverse slides is an open question. Independent benchmarks may suffer from data leakage with public datasets. The computational cost of the largest models limits accessibility, though models trained on far fewer WSIs can match Virchow2 [136]. Bidirectional external validation between patch benchmarks and calibration-focused evaluation when the test domain changes are also seldom reported as primary endpoints. Recent reviews note that task-specific AI models depend on large annotated training sets, which is a rate-limiting factor for pathology AI development, and that even foundation-model pipelines still face clinical integration and trust barriers [137] Finally, the optimal combination of adaptive preprocessing, domain-specific augmentation, and foundation model transfer learning for cross-centre generalizability, the central aim of the present work, has not been systematically investigated.

Chapter 4 Methodology

This chapter covers patch-level binary tumour classification on (H&E) patches from PCam and CAMELYON17 (WILDS). Patches were screened for within-dataset duplicates, cross-dataset overlap, and image quality; those that failed were removed, and the remainder was filtered and stain-normalized with single-reference Macenko after a short benchmark of common stain methods. A CNN baseline and a Virchow2 classifier were trained, the latter with a frozen pretrained encoder and an optimized linear head; hyperparameters, the saved checkpoint, and a calibration temperature were chosen from validation data only.

Each fitted model was scored on the held-out test split of its training dataset and then on the other dataset without further training, using one decision threshold and report layout. The mirrored layout is used so that transfer can be compared in both directions; interpretively, the principal external question is how well PCam-trained weights perform on multi-hospital CAMELYON17 test, while the reverse direction supplies prespecified information on asymmetry and on training from heterogeneous nodal data before testing on PCam. Discrimination, calibration, deterministic test-time probabilities for each patch, and a small structured qualitative review were combined with bootstrap intervals for the main metrics, paired permutation tests for planned model comparisons on the same test material, and a fixed ordering when several hypotheses were reported together. The overall workflow is shown in Figure 4.1.

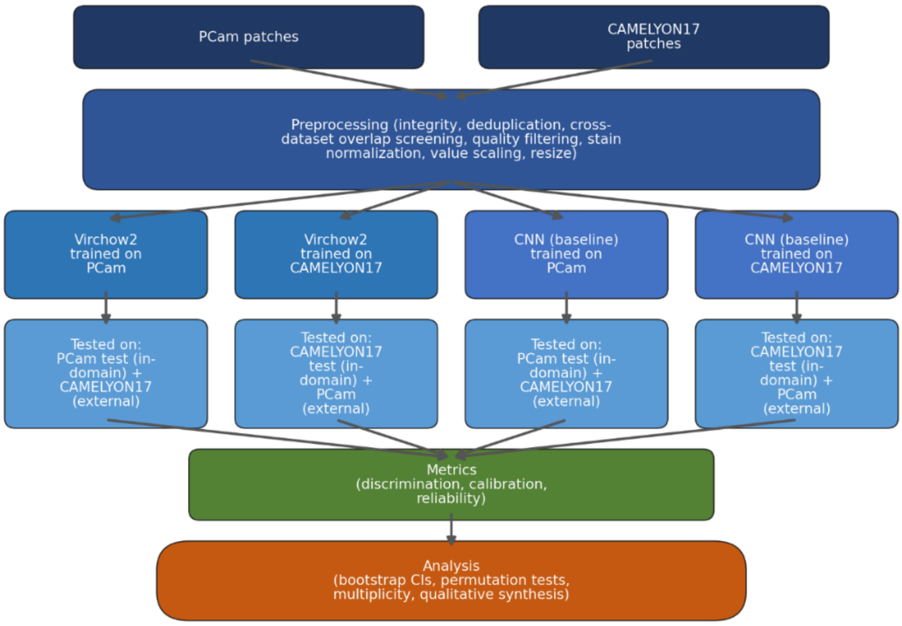

Figure 4.1. Schematic of the study workflow.

4.1 Study Design

The study is retrospective and uses only PCam and CAMELYON17 (WILDS). For each architecture, models are trained on one dataset, scored on its official held-out test split, then scored on the other dataset without retraining, under a single reporting protocol; the same layout is repeated with train and external roles swapped so transfer asymmetry is visible under a prespecified mirror. It is not assumed that both directions must meet the same bar: the principal interpretive question is how well PCam-trained weights perform on multi-hospital CAMELYON17 test (with in-domain PCam test as reference), i.e. whether the integrity-aware pipeline supports deployment-style transfer from a large, curated corpus to a harder external nodal benchmark. The mirror direction (train on CAMELYON17, test on PCam) is reported in full for discrimination, calibration, and errors to characterize asymmetry, including cases where ranking stays strong while recall at a fixed threshold and calibration degrade, and is not treated as a symmetric second success criterion for the forward claim. Hyperparameters, the saved checkpoint, and post-hoc temperature scaling use only the training corpus's validation split; test labels from neither corpus steer those choices.

Primary endpoints are discriminative performance on the non-training test surface and the drop from in-domain test (ROC-AUC and PR-AUC, with absolute or relative change where helpful).

Threshold-based behaviour after calibration is reported at probability 0.5 (accuracy, recall, F1). Secondary endpoints are calibration stability when the test domain changes and agreement of a small structured qualitative review with tabular patterns.

4.2 Datasets

PatchCamelyon and CAMELYON17 as packaged in WILDS were the only data sources. Both address the same clinical problem class: detection of lymph node metastasis in (H&E) sections from the sentinel or axillary lymph node assessment pathway in breast cancer, where the task is to find metastatic tumour in nodal tissue on H&E rather than to classify a primary breast tumour on its own. PCam was built by extracting fixed (96×96) RGB patches from whole-slide images released with CAMELYON16. CAMELYON17 follows the CAMELYON17 challenge lineage; in the WILDS release, patches are (96×96) H&E crops from lymph node slides collected at several Dutch hospitals, with hospital identifiers supplied as metadata so that inter-site differences in staining and acquisition are explicit, while the label remains patch-level metastasis detection in the same nodal setting. For both benchmarks, the public definition of a positive patch is aligned: positivity requires tumour tissue in the central (32×32) pixel region of each (96×96) field, with the outer rim providing context only. PCam was used as a large development benchmark and in-domain reference; CAMELYON17 was used as the principal external target when training on PCam, because its test split aggregates multi-hospital nodal tissue and stresses appearance relative to PCam. The transfer layout was fixed and mirrored as in Table 4.1. In-domain testing was defined as evaluation on the held-out test split of the dataset on which the model was trained; external-domain testing was defined as evaluation on the other dataset with weights frozen after training and calibration on the source domain.

Table 4.1. Definition of development, in-domain test, and external-domain test sets for bidirectional transfer experiments.

Direction	Development Set (Train + Validation)	In-Domain Test Set	External-Domain Test Set
PCam to CAMELYON17	PCam	PCam test split	CAMELYON17
CAMELYON17 to PCam	CAMELYON17	CAMELYON17 test split	PCam

Before any model was trained, split-level descriptive statistics were computed so that class balance and coarse appearance could be compared without fitted parameters. Each quantity was evaluated per patch and then summarized within train, validation, and test for PCam and for CAMELYON17. For every split, 4,000 patches were drawn at random with a fixed sampling design; the same sample size and the same descriptor set were used in both datasets. The descriptor list is given in Table 4.2.

Table 4.2 Summary of split-level statistical descriptors used for pre-model dataset characterization

Category	Descriptors
Class composition	Number of positives, number of negatives, positive fraction
Channel and grayscale appearance	Mean and standard deviation of R, G, B, and grayscale intensity
Colour and tissue proxies	Saturation-based tissue-content proxy, black-pixel ratio, blue-dominance ratio, pink-proxy ratio
Intensity bounds	Patch-level minimum and maximum intensity
Split-level aggregation	Mean, standard deviation, minimum, and maximum across sampled patches

4.3 Preprocessing

The preprocessing stage was meant to eliminate data leakage and reduce non-biological variation while keeping the patch definition aligned with the public benchmarks. Train, validation, and test were handled under fixed rules, PCam was deduplicated by pixel hashing, CAMELYON17 and PCam were checked for exact cross-dataset overlap, and hyperparameter search together with temperature scaling were restricted to source validation so that test material never steered those choices. After those integrity steps, patches that failed simple image-quality gates (solid-colour, high-black, or low-tissue behaviour) were dropped so that stain normalization and training did not spend capacity on empty or artifact-dominated tiles. Stain normalization was then applied to align H&E appearance across scanners and sites while leaving diagnostically relevant structure in place, followed by linear rescaling of stored uint8 RGB to $[0,1]$. Patches stayed on disk at native (96×96); for the Virchow2 backbone, bicubic upsampling to (224×224) was applied only when tensors were formed for the model.

4.3.1 Split Policy and Integrity

Train, validation, and test were kept disjoint within each source. Hyperparameter search and temperature fitting were limited to the source validation split only, and neither in-domain nor external test labels were used for those decisions. Duplicate patches within PCam, hash-based overlap checks between PCam and CAMELYON17, and eligibility for quality-based exclusion were handled under fixed rules before stain normalization. These control rules are summarized in Table 4.3.

Table 4.3 Summary of data integrity and leakage-prevention controls applied in the study

Control Area	Rule Applied
Split isolation	Train, validation, and test splits were treated as disjoint sets within each dataset.
Model selection boundary	Hyperparameter tuning and calibration were restricted to source-domain validation data.
Test-set protection	In-domain and external-domain test sets were not used for model selection.
Duplicate control (within PCam)	Exact duplicate patches were screened by SHA-256 hashing; one index per duplicate group was retained.
Cross-dataset overlap screening	Exact hash matching was run between CAMELYON17 and PCam to identify potential pixel-level overlap.
Quality-control exclusions	Low-quality patches were removed using predefined criteria (solid-colour, high-black, and low-tissue filters).

4.3.2 Quality Filtering

Quality filtering was applied after the split and integrity steps to exclude patches with limited histologic usefulness before stain normalization and model training. The objective was to reduce non-informative inputs (for example, low-texture regions, very dark artifacts, and tissue-poor patches) that can bias optimisation and inflate variance without adding diagnostic evidence. In practical terms, this step prioritizes tissue-bearing patches with interpretable morphology and consistent colour structure. It does not attempt lesion-level semantic filtering;

instead, it provides a conservative image-quality gate so that downstream learning focuses on biologically meaningful content.

For each patch $I \in [0,1]^{H\times W\times 3}$, grayscale intensity was defined as:

$$Y = 0.299R + 0.587G + 0.114B \tag{19}$$

Pixel-wise saturation was defined as:

$$S = \frac{max(R,G,B) - min(R,G,B)}{max(R,G,B) + \epsilon} \tag{20}$$

where ϵ is a small constant to avoid division by zero.

The final tissue proxy was:

$$TissueRatio = \frac{1}{HW}\sum 1(S > 0.12) \tag{21}$$

A patch was removed if its grayscale variability was below the solid-colour threshold ($\sigma_y < 0.04$), if at least half of its pixels were very dark ($\frac{1}{HW}\sum 1(Y < 0.05) \geq 0.5$), or if the tissue proxy indicated too little stained tissue ($TissueRatio < 0.35$). These thresholds were fixed before final runs and applied consistently across splits.

4.3.3 Stain Normalization

Stain normalization was implemented to reduce colour variability unrelated to tissue biology, particularly variability introduced by differences in staining conditions and acquisition environments. The goal was not to force identical visual appearance across all patches, but to improve colour consistency while preserving morphology that is relevant for diagnosis.

A benchmark of the classical and adaptive stain normalization techniques was conducted in a controlled subset-based setting to compare stain-handling strategies under matched data, architecture, and optimisation conditions. The subsets were created from PCam with class-balanced sampling targets before quality filtering, with train target size 40,000 patches, validation target size 8,000 patches, and test target size 16,000 patches. A shared quality filter was then applied, and the same filtered indices were reused across stain-method variants to ensure a fair method comparison. The stain normalization techniques included in the benchmark were:

- Macenko (classical single-reference)

- Reinhard (classical single-reference)
- Vahadane (classical single-reference)
- Adaptive single-reference routing
- Adaptive multi-reference routing
- Adaptive multi-reference with training-time augmentation

Single-reference methods map every patch to one fixed reference look. The reference came from quality-filtered PCam train tiles with a stricter tissue rule ($T_i \geq 0.50$). Among candidates, the final reference was the tile with smallest weighted distance between its RGB means and tissue ratio and the target profile. That distance is calculated as:

$$d_i = \sqrt{\omega_R\left(\mu_{R,i} - \mu_R^*\right)^2 + \omega_G\left(\mu_{G,i} - \mu_G^*\right)^2 + \omega_B\left(\mu_{B,i} - \mu_B^*\right)^2 + \omega_T(T_i - T^*)^2} \quad (22)$$

where $\mu_{R,i}, \mu_{G,i}, \mu_{B,i}$ are candidate mean channel intensities, T_i is candidate tissue ratio, and $\mu_R^*, \mu_G^*, \mu_B^*, T^*$ is the target profile. Fixed weights were $\omega_R = \omega_B = 1.5$ and $\omega_G = \omega_T = 1.0$, emphasizing R and B over G and tissue fraction. Targets were set for a typical pink–blue H&E balance (R at the 62nd percentile and B at the 38th percentile of the candidate pool). The reference was then reused for all benchmark runs and both datasets.

Multi-reference methods cluster candidates in stain-feature space, pick one central reference per cluster, fit a normalizer for each reference, and assign each patch to its nearest cluster so it is normalized with the matching “reference pack.” Adaptive routing picks a path per patch using blue dominance,

$$BlueDom(I) = \frac{1}{HW}\sum 1(B > R) \quad (23)$$

Per split, τ_{blue} was the 25th percentile of BlueDom on kept patches: tiles with ($BlueDom(I) < \tau_{blue}$) tried Reinhard first, others tried Macenko first; the other method was used if the first failed, and luminosity-only output if both failed. Extra post-normalization checks could also send failing tiles to luminosity-only. Adaptive runs additionally applied a “purple-tail” safeguard: per split, cutoffs from the 2nd percentile of mean red and of pink-ratio sent tiles below either cutoff to luminosity-only. The adaptive multi-reference + augmentation condition kept this routing and added train-only colour and intensity jitter (no geometric warps): gain in $[0.9, 1.1]$, bias in $[-0.03, 0.03]$, saturation scale in $[0.9, 1.1]$, then clip to $[0,1]$; validation and test were not augmented.

For colour-distance monitoring, the mean RGB vector of each patch,

$$\mu(I) = [\mu_R, \mu_G, \mu_B\,] \tag{24}$$

was compared with the reference mean μ_{ref} using Euclidean distance:

$$d(I, I_{ref}) = \left\| \mu(I) - \mu_{ref} \right\|_2 \tag{25}$$

Distance reduction before versus after normalization was used as a quality indicator for stain alignment. In addition to this distance-based check, preprocessing reports were used to monitor per-split behaviour and verify that normalization did not create unstable colour artifacts.

All stain variants used the same baseline CNN and training: four convolutional blocks (32, 64, 128, 256 channels), global average pooling, a 128-unit dropout head, binary cross-entropy with logits, and Adam with shared hyperparameters. Scores were reported with the same metric set (ROC-AUC, accuracy, balanced accuracy, precision, recall, specificity, F1, MCC, and TP/TN/FP/FN). Macenko was chosen for the main experiments because it was the most stable under the cross-domain preprocessing protocol, so final PCam and CAMELYON17 runs used Macenko in the benchmark-style single-reference configuration with that fixed reference.

4.3.4 Value Normalization and Resizing

After quality and stain processing, pixel values were normalized to:

$$I_{norm} = \frac{I}{255} \tag{26}$$

That convention was applied before model input and metric computation so all scores used the same intensity scale. Patch storage remained at 96×96. For architectures requiring 224×224 input, interpolation was applied as:

$$I_{224} = Resize_{bicubic}(I_{96}) \tag{27}$$

Bicubic was chosen over bilinear after side-by-side inspection of representative preprocessed validation patches upsampled to (224×224). Bicubic crops were clearly sharper at nuclear membranes and chromatin texture; bilinear crops looked slightly smoother. Informal timing checks showed no meaningful batch-time penalty for bicubic. Bicubic was therefore adopted for all Virchow2 training and evaluation. Stored patches remain (96×96) crops; upsampling does not restore true high-resolution structure from the source scanner but reduces smoothing before the frozen encoder relative to bilinear interpolation.

4.4 Models and Training

Two model families were used under a harmonized training and evaluation framework: a conventional CNN baseline and a Virchow2-based classifier. The CNN baseline is a compact convolutional network trained end-to-end from random initialization on (96×96) patches. It serves as a conventional, dataset-specific reference without a pretrained pathology backbone. Virchow2 is a large pathology foundation model pretrained on broad whole-slide histopathology data; here its encoder was kept frozen and only a small linear head was trained on top of its patch embeddings (after bicubic upsampling to (224×224). This dual-model design supports both methodological benchmarking and transfer-focused analysis under matched preprocessing and split policies. The experiments run during this study are organized into labelled conditions (codes C1–C8 in Table 4.4). Each condition states where the model trains, which preprocessing and stain arm apply for that row, which architecture is used, and which splits are scored. Within a comparison family, the intent is that only one major factor differs across runs (typically the training domain, the stain policy, or the backbone). Conditions C1–C4 form the main Virchow2 block: in-domain and external tests in both transfer directions under the adopted preprocessing pipeline. Conditions C5–C6 run the stain-method benchmark on the controlled PCam subsets with the shared baseline CNN, varying classical versus adaptive stain handling. Condition C7 is the shallow CNN transfer reference in four training arms (PCam or CAMELYON17 × raw uint8 or study-pipeline tensors), each scored on in-domain and external test with the same mirrored layout as C1–C4. Condition C8 is the structured qualitative error review (fixed checklist, bucket sampling) on Virchow2 outputs for C1–C4 only.

Table 4.4. Experimental condition matrix used to evaluate in-domain performance, cross-domain transfer, stain-handling ablations, and qualitative failure patterns across PCam and CAMELYON17.

Condition ID	Condition Purpose	Stain Policy	Model	Train Dataset	Test Dataset(s)
C1	In-domain PCam baseline evaluation	Macenko benchmark-style	Virchow2 + linear head	PCam	PCam test
C2	External transfer PCam to CAMELYON17	Macenko benchmark-style	Virchow2 + linear head	PCam	CAMELYON17

C3	In-domain CAMELYON17 evaluation	Macenko benchmark-style	Virchow2 + linear head	CAMELYON17	CAMELYON17 test
C4	External transfer CAMELYON17 to PCam	Macenko benchmark-style	Virchow2 + linear head	CAMELYON17	PCam test
C5	Stain-method benchmark (classical single-reference)	Macenko / Reinhard / Vahadane	Baseline CNN	PCam benchmark subset	PCam benchmark test subset
C6	Stain-method benchmark (adaptive variants)	Adaptive single-ref / adaptive multi-ref / adaptive multi-ref + aug	Baseline CNN	PCam benchmark subset	PCam benchmark test subset
C7	Shallow CNN transfer reference (raw and study-pipeline)	Raw uint8 or study's preprocessing pipeline	Baseline CNN	PCam and CAMELYON17 (four arms)	In-domain + external test per arm
C8	Structured qualitative error review	Study's preprocessing pipeline	Virchow2 (C1-C4)	Per condition (C1-C4)	Sampled error buckets on full test splits

4.4.1 CNN Baseline

The CNN baseline served as the conventional reference architecture for both controlled stain-method benchmarking and non-foundation transfer comparison. Its role in the study was to provide a lower-complexity model family against which the behaviour of Virchow2-based transfer could be interpreted under matched preprocessing and evaluation conditions. The architecture consisted of four convolutional stages with channel progression (32, 64, 128, 256), followed by global average pooling and a dense head with 128 hidden units, dropout regularization, and a final binary logit output. Nonlinear activation and downsampling were applied through the feature-extraction pipeline to progressively increase representational abstraction while controlling spatial resolution and parameter growth. All CNN parameters were trainable, and each run started from random initialization. Optimisation used binary cross-entropy with logits and Adam under fixed settings within each comparison family. This configuration allowed stain-handling effects to be compared without confounding from architecture changes or inconsistent optimisation rules.

4.4.2 Virchow2 Classifier

The primary model family for cross-domain analysis was a Virchow2-based binary classifier. In this setup, each input patch was first mapped by the pretrained Virchow2 encoder into a high-level pathology representation, and this representation was then passed to a linear classification head that produced the tumour-versus-non-tumour logit. The trainable-parameter policy followed a frozen-feature protocol across both transfer directions. Encoder weights were kept fixed, and only the linear head was optimized during supervised training. This design made the adaptation stage computationally tractable, reduced instability risk during optimisation, and limited overfitting pressure in cross-domain settings where distribution mismatch is expected. The Virchow2 head was trained with the same loss formulation, optimizer family, and validation logic used in the shared study framework, so that differences between model families could be interpreted as representation and transfer effects rather than differences in development procedure. Keeping the encoder frozen also separated preprocessing-domain effects from deeper representation reconfiguration.

4.4.3 Training and Model Selection

To ensure model-family comparisons remained interpretable, training was harmonized across CNN and Virchow2 conditions by fixing the development protocol within each experiment family. For the full-data transfer experiments, the conventional CNN baseline and the Virchow2 linear head were each trained for ten epochs so the number of passes over the training indices matched; the CNN still applies a full-network gradient step each iteration whereas Virchow2 updates only the head on frozen encoder features, so per-step compute and effective capacity are not identical even though the optimisation horizon is aligned. Non-target settings (including the full preprocessing order from split integrity through resizing, the optimisation framework, and the reporting template) were kept consistent so that observed differences could be attributed to the intended ablation factor. Model selection and hyperparameter control remained validation-bound at the source domain, with no target-domain information used during development. Reproducibility was supported through explicit seed control and run-level artifact logging of configurations and outputs. Checkpointing followed a validation-driven policy. For each run, model states were monitored during training and the checkpoint achieving the best source-domain validation performance under the predefined primary selection criterion was retained as the final reporting model. The selection rules differed by experiment family so that each matched how discrimination was optimized in that block: the shallow CNN on full-transfer arms and on stain-benchmark subsets used the epoch with highest validation ROC-AUC on raw sigmoid probabilities; Virchow2 used the epoch with highest validation accuracy on the source validation split; the stain-method benchmark used validation ROC-AUC for checkpointing and test ROC-AUC on the held-out benchmark split to choose Macenko for full-corpus runs. Held-out source-domain test evaluation and external-domain test evaluation were executed only after checkpoint selection had been finalized. This ordering prevented leakage of test information into model development and maintained consistency across in-domain and transfer comparisons.

4.5 Outcomes

Each model was evaluated in-domain on the source test split and externally on the other dataset without retraining or target-specific tuning. PCam-trained models were scored on PCam test and on CAMELYON17; CAMELYON17-trained models were scored on CAMELYON17 test and on PCam test, under the same protocol, so that the forward external readout (PCam→CAMELYON17) sits alongside a prespecified mirror for asymmetry and calibration behaviour.

Transfer degradation compares each metric's in-domain and external-domain value. For metrics where higher is better (for example ROC-AUC, PR-AUC, accuracy, and F1), M_{in} and M_{ext} are denoted as the in-domain and external-domain values, respectively, and the transfer degradation is computed as:

$$\Delta M_{abs} = M_{in} - M_{ext} \tag{28}$$

$$\Delta M_{rel} = \frac{M_{in} - M_{ext}}{M_{in} + \epsilon} \tag{29}$$

where ϵ is a small positive constant for numerical stability. For metrics where lower is better (for example log loss and Brier score), transfer degradation was summarized by the external minus in-domain change,

$$\Delta L_{abs} = L_{ext} - L_{in} \tag{30}$$

The same sign convention links discrimination and reliability summaries.

Discrimination used ROC-AUC and PR-AUC; threshold-based reporting used accuracy, precision, recall (sensitivity), specificity, F1, confusion-matrix counts (TP, TN, FP, FN), and MCC for imbalance-aware correlation. Probability quality used Brier score, log loss, and ECE.

4.5.1 Calibration

Binary decisions used a fixed probability cut-off of 0.5 on the reported (calibrated) positive probability $\hat{p}$, with no per-domain retuning of the cut-off. The resulting class label is

$$\hat{y} = 1(\hat{p} \geq 0.5) \tag{31}$$

with $\hat{y} \in \{0,1\}$ and $1(\cdot)$ the indicator function. The same rule was applied on source test and external test for every condition, so transfer comparisons are not confounded by post hoc threshold choice.

Temperature scaling maps each pre-calibration logit $z(x)$ through a single scalar $T > 0$ learned on source-domain validation only. Calibrated probabilities are

$$\hat{p}_T(y = 1 \mid x) = \sigma\left(\frac{z(x)}{T}\right) \tag{32}$$

with $\sigma(\cdot)$ the logistic sigmoid; that T was then applied unchanged to source test and external test. External labels and logits were not used to fit or adjust T, in line with the split rules in Section 4.3.1 Split Policy and Integrity. Brier score, log loss, and ECE summarize how well those probabilities match observed outcomes under this mapping.

4.5.2 Deterministic test scores and error subsets

Main tables and figures use one deterministic forward pass per patch (head dropout off), followed by the sigmoid and the same temperature mapping when calibrated probabilities are reported. From the resulting $\hat{p}(x)$, case-level summaries use confidence $c(x) = \max(\hat{p}(x), 1 - \hat{p}(x))$ and predictive entropy:

$$H(x) = -\bar{p}(x) \log(\bar{p}(x)) - \big(1 - \bar{p}(x)\big) \log\big(1 - \bar{p}(x)\big) \tag{33}$$

High $H(x)$, with low $c(x)$ is treated only as a descriptive flag for ambiguous single-pass predictions, not as a full uncertainty model.

Failure sets were fixed before inspection. With ground truth y and predicted class $\hat{y}$, misclassified patches satisfy $\hat{y} \neq y$. Two reliability-focused subsets on errors are

$$\varepsilon_{conf} = \{x: \hat{y}(x) \neq y(x), c(x) \geq \tau_c\} \quad (34)$$

$$\varepsilon_{unc} = \{x: \hat{y}(x) \neq y(x), H(x) \geq \tau_H\} \quad (35)$$

with $\tau_c = 0.9$ and τ_H the 90th percentile of $H(x)$ within each evaluation split, so confident wrong calls can be separated from wrong calls under flatter probabilities. The same cut-offs, calibration rule, and definitions were used in both transfer directions (PCam to CAMELYON17 and CAMELYON17 to PCam).

4.6 Qualitative analysis

A structured visual review was done on the same deterministic test outputs summarized in Section 4.5, to relate recurring patch appearance and quality-related cues to the transfer and calibration results. It supplements the tables rather than replacing them. Buckets and sampling rules were fixed beforehand, so reviewed material stays comparable across transfer directions and model conditions. Figure 4.2 outlines how full-test predictions are split into error and uncertainty pools, how patches are sampled from those pools for export, and how structured checklist review relates to, but does not replace, aggregate metrics.

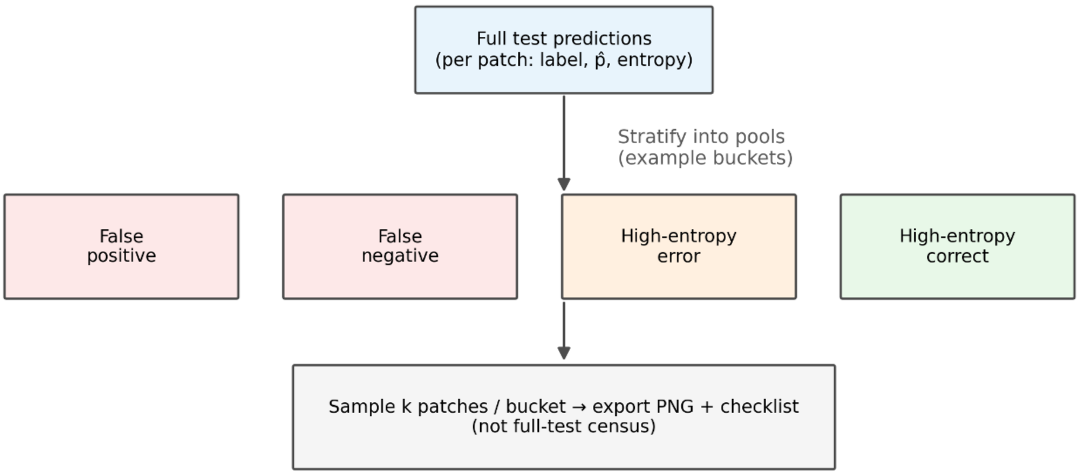

Figure 4.2. Qualitative review buckets and sampling.

Buckets use the same y, $\hat{y}$, $c(x)$, $H(x)$, and $\hat{p}(x)$ as in Section 4.5.2 Deterministic test scores and error subsets. With τ_H the 90th percentile of $H(x)$ within each evaluation split, the primary sets are

$$\mathcal{B}_{FP} = \{x: \hat{y}(x) = 1, y(x) = 0\} \quad (36)$$

$$\mathcal{B}_{FN} = \{x: \hat{y}(x) = 0, y(x) = 1\} \tag{37}$$

$$\mathcal{B}_{UE} = \{x: \hat{y}(x) \neq y(x), H(x) \geq \tau_H\} \tag{38}$$

$$\mathcal{B}_{UC} = \{x: \hat{y}(x) = y(x), H(x) \geq \tau_H\} \tag{39}$$

and high-confidence errors are tracked separately as

$$\mathcal{B}_{CE} = \{x: \hat{y}(x) \neq y(x), c(x) \geq 0.9\} \tag{40}$$

For each transfer direction the review targeted 80 cases in total (up to 20 each from false positives, false negatives, high-entropy mistakes, and high-entropy correct calls) with random draws within a bucket using a fixed seed. Buckets smaller than the target contributed all available patches and the shortfall was noted in reporting. Tabulated qualitative prevalences in the results chapter refer to specific buckets only (for example false negatives and high-entropy errors at $k/10$ per condition); they must not be read as if every exported case or every bucket contributed to those tables.

Each case was stored in a single evidence package: raw patch, preprocessed patch, label, prediction, $\hat{p}$, c, H, bucket, transfer direction, and model condition, so visuals and numeric context were read together. Review used a fixed checklist with Present, Absent, or Unclear for tissue scarcity, artifact burden, borderline morphology, small-focus lesion pattern, colour or stain atypia, and patch-context limits. The same rubric and randomized within-bucket order were used in every bucket and direction to limit reviewer drift.

Within each bucket (b) and checklist pattern (r), prevalence was summarized by

$$\hat{p}_{r,b} = \frac{k_{r,b}}{n_b} \tag{41}$$

where $k_{r,b}$ is the number of reviewed cases marked Present for pattern r, and n_b is the number of reviewed cases in bucket b. Those counts were read next to small representative panels when figures were prepared. Qualitative counts support interpretation of failure modes and domain-sensitive errors alongside Section 4.5; they are not treated as free-standing proof of causality, and claims stay tied to what was actually seen in the sampled set.

4.7 Statistical analysis

Case-level bootstrap with $B = 2000$ replicates was used to attach 95% percentile intervals to the point metrics in Section 4.5, and paired permutation tests were used when two methods were compared on the same test patches. Primary versus secondary claims and multiplicity rules were fixed before the results were read, to keep confirmatory statements from drifting with post hoc choices. Figure 4.3 distinguishes how bootstrap resampling and paired permutation tests reuse the fixed test set: the former builds a sampling distribution for a single model's metric; the latter compares two models on identical patch indices under an exchangeability null.

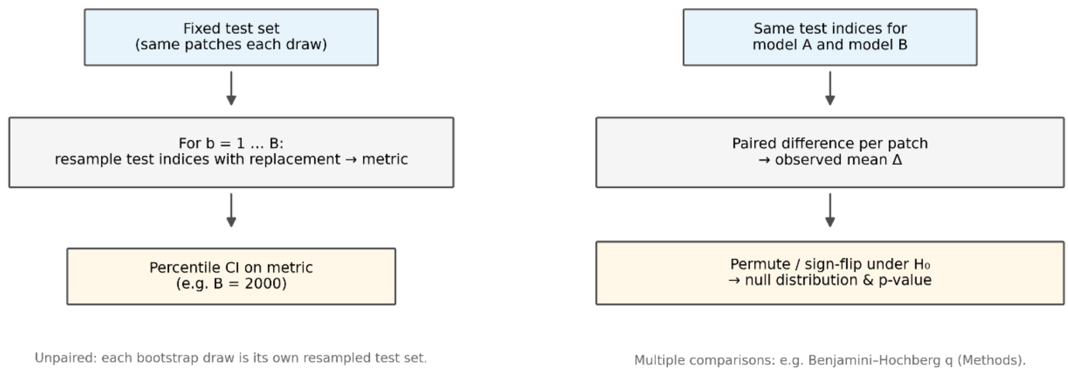

Figure 4.3. Bootstrap intervals vs paired permutation.

Primary inference targeted the main Virchow2 pipeline: within each transfer direction, the drop from in-domain to external performance, and across directions, asymmetry in those drops. Secondary material (for example model-family contrasts or qualitative prevalence summaries) was read as supportive unless it still cleared multiplicity control. On a test split of size n, each scalar metric M was summarized by a point estimate on the observed cases.

$$\widehat{M} = M(\{y_i, \hat{p}_i\}_{i=1}^{n}) \tag{42}$$

Bootstrap replicate b drew a case-level resample with replacement and recomputed the metric on that replicate.

$$\widehat{M}^{(b)} = M\left(\left\{y_i^{*(b)}, \hat{p}_i^{*(b)}\right\}_{i=1}^{n}\right) \tag{43}$$

The 95% interval was the 2.5th and 97.5th percentiles of the bootstrap distribution $\{\widehat{M}^{(1)}, \dots, \widehat{M}^{(B)}\}$:

$$CI_{95\%}(M) = \left[Q_{0.025}\left(\widehat{M}^{(1:B)}\right), Q_{0.975}\left(\widehat{M}^{(1:B)}\right)\right] \tag{44}$$

The same bootstrap machinery was applied to discrimination, threshold-based, and reliability metrics. Transfer drops followed Section 4.5.1 Calibration, using bootstrap point estimates $\widehat{M}_{in}$ and $\widehat{M}_{ext}$ within each replicate.

$$\Delta M_{abs} = \widehat{M}_{in} - \widehat{M}_{ext} \tag{45}$$

$$\Delta M_{rel} = \frac{\widehat{M}_{in} - \widehat{M}_{ext}}{\widehat{M}_{in} + \epsilon} \tag{46}$$

Intervals for those drop quantities came from the paired bootstrap draws of in-domain and external estimates within each replicate. Head-to-head model comparisons on a shared test set used paired permutation tests on case-level score differences d_i (for example squared-error pieces for Brier or log-loss pieces for NLL). The tested null was

$$H_0: \mathbb{E}[d_i] = 0 \tag{47}$$

Permutation p-values used 10,000 random sign flips of d_i so cases stayed matched. Where a metric did not admit a stable per-case split, reporting relied on bootstrap interval overlap and effect size, and that limitation was stated in the results. Confirmatory p-values were limited to the primary claim family and adjusted across those tests by Benjamini–Hochberg at $q = 0.05$. Other analyses kept unadjusted intervals and p-values and were labelled exploratory. Tables emphasized effect size and interval width, with adjusted or unadjusted p-values by tier and a short robustness interpretation rather than p-values alone. For qualitative prevalence summaries (Section 4.6), sample prevalence was written

$$\hat{p} = \frac{k}{n} \tag{48}$$

with intervals when n allowed it; otherwise, raw counts were emphasized.

Chapter 5 Results and Discussion

This chapter presents the empirical outcomes of the study. It first characterizes the PCam and CAMELYON17 corpora after deduplication, overlap screening, and quality control, then compares preprocessing choices through targeted ablations, including stain-normalization strategies, input upsampling for the foundation-model pipeline, and the effect of the full study preprocessing pipeline on a shallow baseline trained on raw public patches versus curated tensors. Classifier performance is reported for a convolutional baseline and for Virchow2 under the same cross-domain protocol: each model is trained on one dataset, evaluated on its in-domain test split, and evaluated on the other dataset without retraining. The interpretive emphasis is external transfer when training on PCam and testing on held-out multi-hospital CAMELYON17 test—the direction that motivated the integrity-aware pipeline—while the mirrored direction (train on CAMELYON17, test on PCam) is reported in full to document asymmetry and how optimizing on heterogeneous nodal data interacts with PCam's distribution; it is not treated as a second symmetric bar for the forward claim. Discrimination, calibration, and uncertainty-related summaries are given together with bootstrap intervals and paired tests where two models or training origins share the same test patches. A structured qualitative review of sampled errors is included to relate numeric transfer patterns to visible tissue, stain, and context factors. The discussion that follows interprets how preprocessing, model capacity, and domain shift interact, rather than restating the experimental protocol.

5.1 Datasets

PCam and CAMELYON17 were used as fixed 96×96 RGB patch benchmarks with standard train, validation, and test partitions. Table 5.1 lists catalogue sizes before study-specific curation. Only PCam underwent exact-content deduplication; CAMELYON17 counts refer to the indexed WILDS pack before quality filtering.

Table 5.1 Patch counts per split before quality filtering and stain normalization.

Split	PCam	CAMELYON17
Training	262,144	302,436
Validation	32,768	34,904
Test	32,768	85,054
All splits	327,680	422,394

After deduplication, PCam retained 277,516 unique patches across splits (50,164 duplicate indices removed). Table 5.2 gives the split-wise reduction. CAMELYON17 was not deduplicated at the patch level; its split sizes in Table 5.1 apply through to the quality-filter stage.

Table 5.2 PCam counts before and after exact-content deduplication.

Split	Before	After	Removed
Training	262,144	220,025	42,119
Validation	32,768	28,108	4,660
Test	32,768	29,383	3,385
All splits	**327,680**	**277,516**	**50,164**

Cross-dataset leakage was assessed by SHA-256 hashing of uint8 RGB tensors on the final preprocessed corpora. Table 5.3 summarizes directional overlap between corpora. At most, four patches in a CAMELYON17 training split matched any PCam hash; rates were on the order of 10^{-5}. Exactly one hash appeared in both pooled unions (262,540 PCam-unique and 413,860 CAMELYON17-unique). Within CAMELYON17 alone, three training pairs were byte-identical duplicates; validation and test contained none. These counts indicate that aggregate test performance is not explained by shared pixels across datasets, although visually similar but non-identical tiles remain possible.

Table 5.3 Directional exact-hash overlap on preprocessed tensors.

Split	CAMELYON17 patches	Overlap with any PCam	PCam patches	Overlap with any CAMELYON17
Training	296,294	4	208,355	2
Validation	34,389	0	26,515	0
Test	83,181	1	27,704	0

5.2 Quality Filtering and Class Balance

Quality screening removed 14,942 of 277,516 PCam candidates (5.4%) and 8,530 of 422,394 CAMELYON17 candidates (2.0%). Low-tissue patches accounted for most removals; high-black tiles were more frequent on CAMELYON17 (886) than on PCam (13). Table 5.4 pools removal reasons. Split-wise throughput and retained class counts appear in Table 5.5 and Table 5.6. Positive fractions among kept patches remained close to pre-filter levels, so the screen did not materially shift label prevalence.

Table 5.4 Pooled quality-filter removals (all splits).

Dataset	Candidates	Retained	Removed	Low tissue	Solid colour	High black
PCam	277,516	262,574	14,942	12,451	2,478	13
CAMELYON17	422,394	413,864	8,530	6,834	810	886

Table 5.5 Quality-filter throughput by train, validation, and test split.

Split	PCam candidates	PCam retained	PCam removed	CAMELYON17 candidates	CAMELYON17 retained	CAMELYON17 removed
Training	220,025	208,355	11,670	302,436	296,294	6,142
Validation	28,108	26,515	1,593	34,904	34,389	515
Test	29,383	27,704	1,679	85,054	83,181	1,873

Table 5.6 Positive and negative label counts among retained patches.

Split	PCam n_+	PCam n_-	PCam f_+	CAMELYON17 n_+	CAMELYON17 n_-	CAMELYON17 f_+
Training	88,828	119,527	0.426	150,867	145,427	0.509
Validation	11,715	14,800	0.442	17,427	16,962	0.507
Test	12,940	14,764	0.467	42,468	40,713	0.511

Figure 5.1 shows representative tiles rejected for low tissue, solid colour, and high black, alongside borderline patches retained under the fixed thresholds. The panels illustrate that most discarded patches are largely acellular or artifact-dominated rather than morphologically ambiguous tumour deposits.

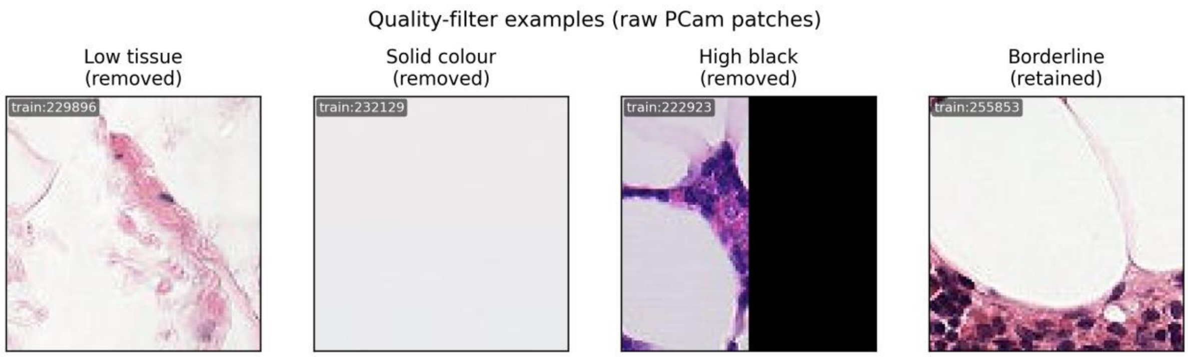

Figure 5.1. Examples of PCam patches removed by the quality filter (low tissue, solid colour, high black) and one borderline patch retained.

5.3 Stain Normalization Benchmark

Six preprocessing arms were compared on class-balanced PCam benchmark subsets with a shared shallow CNN, using the benchmark protocol in the Methodology chapter. Checkpoints were selected by best validation ROC-AUC, and the preprocessing arm for all main runs was chosen by test ROC-AUC on the held-out benchmark split (the discrimination endpoint fixed before test evaluation) because it measures class separability across thresholds and matches the validation criterion, whereas accuracy and F1 at a fixed 0.5 cutoff depend on class balance and an arbitrary operating point and were reported only as secondary threshold metrics. Figure 5.2 reports validation ROC-AUC, test ROC-AUC, test accuracy, and test F1 for each arm. Macenko (classical single-reference) achieved the highest test ROC-AUC (0.925) and was adopted for full-corpus preprocessing and all subsequent classifier runs. Adaptive single-reference routing yielded the highest test accuracy (0.829) and F1 (0.846) at threshold 0.5 on this architecture but not the highest test ROC-AUC. Reinhard reached the highest validation ROC-AUC yet lower test ROC-AUC, showing that validation-based selection did not preserve test ordering for that arm.

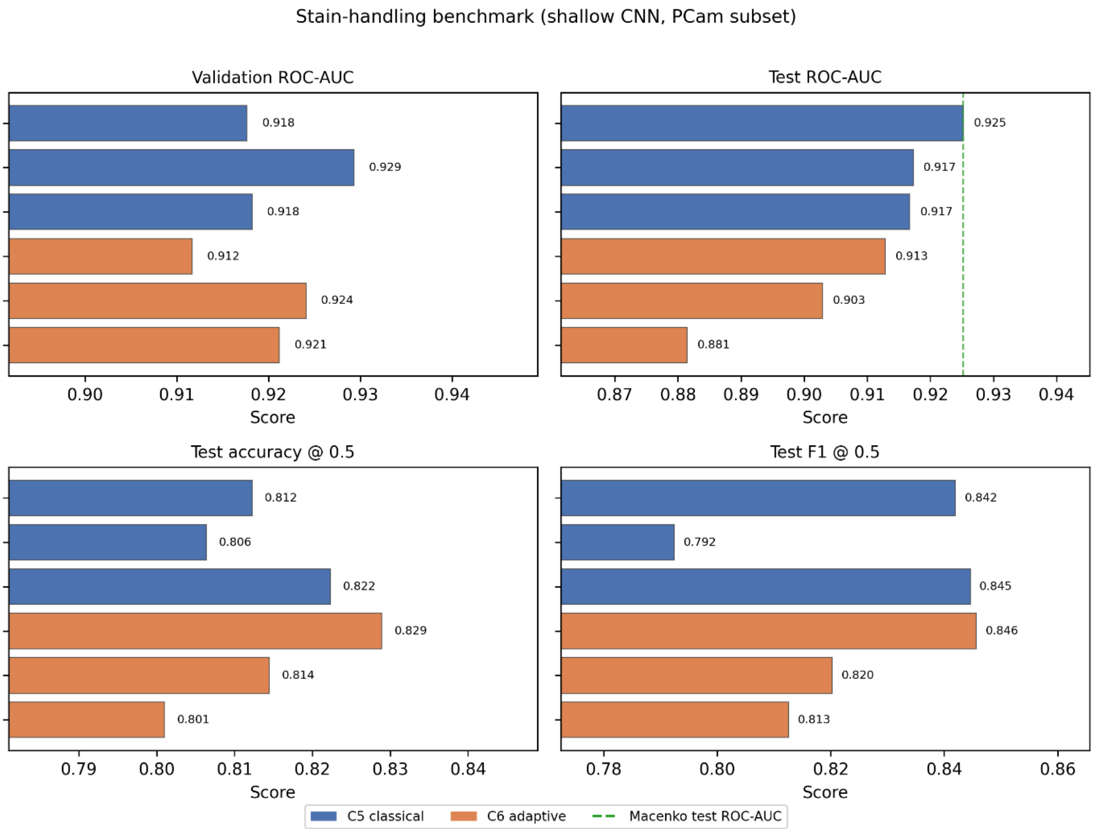

Figure 5.2. Stain-handling benchmark (shallow CNN on class-balanced PCam subsets): validation ROC-AUC, test ROC-AUC, test accuracy, and test F1 for six preprocessing arms (C5 classical, blue; C6 adaptive, orange), ordered by test ROC-AUC. Dashed line: Macenko (selected).

5.4 Descriptive Appearance of Retained Corpora

After quality screening and adoption of Macenko stain normalization for the full corpora, class balance and coarse RGB statistics were summarized on stratified random samples of 4,000 patches per split. Table 5.7 reports sample means for f_+ and appearance descriptors; the n column is the full retained corpus size per split (not the sample size). CAMELYON17 patches showed higher tissue-proxy scores and slightly higher positive fractions than PCam on training and test, consistent with the WILDS pack's label distribution. Blue-dominance ratios differed between datasets, reflecting residual stain and scanner variation that the normalization step was designed to reduce but did not fully erase at the level of simple colour descriptors.

Table 5.7 Descriptive RGB and class-balance statistics on stratified samples after quality control and Macenko normalization.

Dataset	**Split**	n	f_+	**Mean R**	**Mean G**	**Mean B**	**Tissue proxy**	**Black ratio**
PCam	Train	208,355	0.426	0.715	0.579	0.798	0.733	3.7 × 10^(−4)

PCam	Valid	26,515	0.442	0.714	0.576	0.797	0.738	3.6 $\times 10^{\wedge}(-4)$
PCam	Test	27,704	0.467	0.718	0.580	0.799	0.731	4.0 $\times 10^{\wedge}(-4)$
CAMELYON17	Train	296,294	0.509	0.757	0.548	0.816	0.825	2.5 $\times 10^{\wedge}(-4)$
CAMELYON17	Valid	34,389	0.507	0.686	0.490	0.769	0.850	7.3 $\times 10^{\wedge}(-4)$
CAMELYON17	Test	83,181	0.511	0.718	0.514	0.790	0.812	2.3 $\times 10^{\wedge}(-4)$

5.5 Input Resizing

Since Virchow2 expects 224×224 pixel inputs while the stored patches remain at their native resolution, we evaluated different resizing kernels. Side-by-side inspection of validation patches revealed that bicubic crops were clearly sharper and better-preserved nuclear membranes and chromatin textures compared to the slightly smoother, more blurred bilinear crops as shown in Figure 5.3. Additionally, pipeline timing checks showed no meaningful training penalty for bicubic over bilinear at batch time. Consequently, bicubic upsampling was adopted for all Virchow2 training and evaluation based on these quality and performance outcomes.

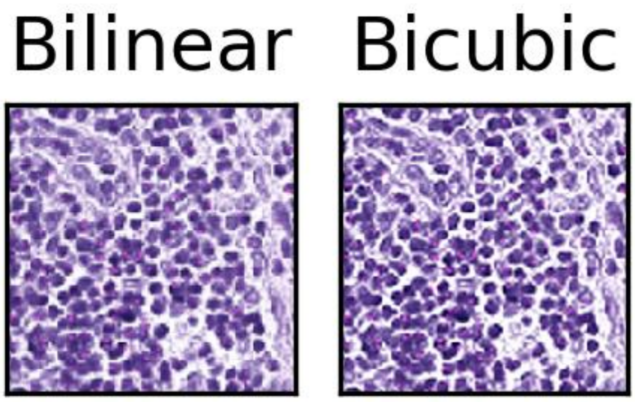

Figure 5.3. The same preprocessed PCam patch upsampled to 224×224 with bilinear (left) and bicubic (right) interpolation. Bilinear smoothing blurs nuclear borders more than bicubic.

5.6 Training Convergence

Each classifier was trained for ten epochs with checkpoints saved every epoch. Figure 5.4 shows per-epoch training loss and training accuracy for the shallow CNN on preprocessed

PCam and CAMELYON17, together with validation ROC-AUC (the metric used to select checkpoints). Figure 5.5 shows training and validation loss and accuracy for Virchow2 with a frozen encoder and trainable linear head on the same corpora. Validation accuracy plateaued after roughly five to seven epochs on PCam validation for both model families; CAMELYON17 validation metrics rose steadily through epoch ten. Best validation checkpoints were carried forward for temperature fitting and test evaluation.

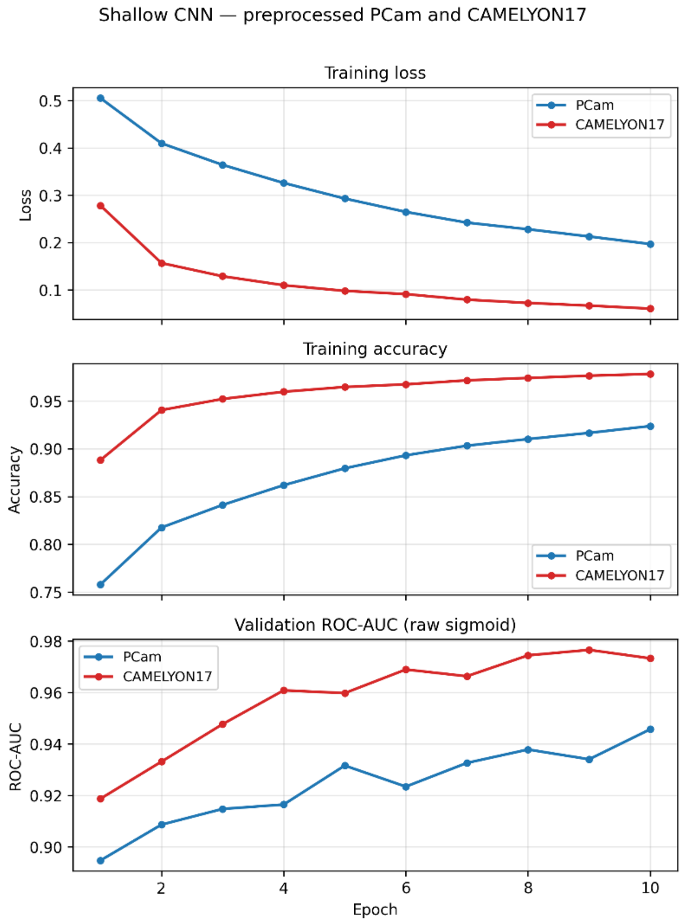

Figure 5.4. Shallow CNN on preprocessed PCam and CAMELYON17: train loss, train accuracy, and validation ROC-AUC (10 epochs; blue/red).

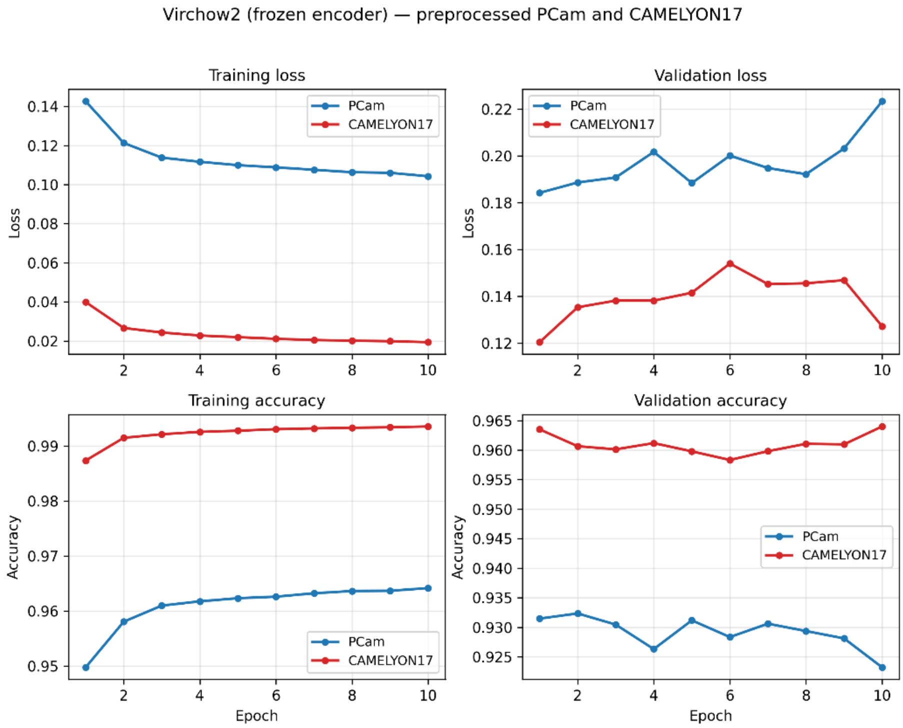

Figure 5.5. Virchow2 (frozen encoder) on the same corpora: train and validation loss and accuracy (10 epochs; blue/red).

5.7 Shallow CNN Baseline

A four-stage convolutional baseline with global average pooling and a dropout head was trained on four arms: PCam and CAMELYON17, each either on raw public patches or on tensors that passed the full study preprocessing pipeline (quality filter, Macenko stain normalization, rescaling to [0,1]). Checkpoints were chosen by best validation ROC-AUC on raw sigmoid probabilities; temperature T was fit on the source validation split only. Results are reported first for raw patches, then for preprocessed patches on the same transfer layout used for Virchow2.

5.7.1 Raw Patches

These arms use catalogue uint8 RGB tensors without any preprocessing. Test splits keep public sizes before the study quality-filter shrink. Table 5.8 shows that CAMELYON17-trained weights on raw in-domain CAMELYON17 test achieved only sub-optimal performance: ROC-AUC 0.724 (above random ranking at 0.50, but well below usable transfer), accuracy 0.554, and recall 0.16 at threshold 0.5, so the model poorly separated classes in practice despite non-trivial ranking. PCam-trained raw weights reached moderate in-domain discrimination (ROC-AUC 0.880). Applying the full preprocessing pipeline was therefore necessary before a fair cross-domain comparison with the main experiments.

Table 5.8. Shallow CNN test performance on raw public patches (calibrated probabilities, threshold 0.5).

Train → Test	n_{test}	ROC-AUC	Accuracy	Recall	Brier	ECE_{15}
PCam → PCam	32,768	0.880	0.780	0.61	0.161	0.139
PCam → CAMELYON17	85,054	0.984	0.936	0.93	0.047	0.013
CAMELYON17 → CAMELYON17	85,054	0.724	0.554	0.16	0.268	0.230
CAMELYON17 → PCam	32,768	0.676	0.521	0.08	0.308	0.280

5.7.2 Preprocessed Patches

Table 5.9 and Table 5.10 give held-out test metrics on quality-filtered, Macenko-normalized tensors scaled to $[0,1]$ ($n_{test} = 27{,}704$ for Pcam test, 83,181 for CAMELYON17 test). The PCam→CAMELYON17 cells are the principal external readout for whether a classifier developed on the large curated corpus still performs on the harder multi-hospital test; the reverse cells quantify the mirrored stress test. Figure 5.6 shows confusion matrices at threshold 0.5 (row-normalized fractions with raw counts annotated). Table 5.11 reports transfer contrasts as $\Delta = M_{in} - M_{ext}$ on calibrated probabilities.

Table 5.9 Shallow CNN test performance (T, ROC-AUC, PR-AUC, accuracy, balanced accuracy) on study-preprocessed patches (calibrated probabilities, threshold 0.5).

Train domain	Test domain	T	ROC-AUC	PR-AUC	Accuracy	Balanced acc.
PCam	PCam	1.473	0.948	0.946	0.871	0.866
PCam	CAMELYON17	1.473	0.985	0.987	0.926	0.926
CAMELYON17	CAMELYON17	1.733	0.985	0.988	0.942	0.943
CAMELYON17	PCam	1.733	0.856	0.867	0.698	0.677

Table 5.10 Shallow CNN test performance (F1, Brier, log loss, ECE15) on study-preprocessed patches (calibrated probabilities, threshold 0.5).

Train domain	Test domain	F1	Brier	Log loss	ECE_{15}
PCam	PCam	0.852	0.094	0.301	0.043
PCam	CAMELYON17	0.931	0.054	0.182	0.058
CAMELYON17	CAMELYON17	0.942	0.045	0.158	0.027
CAMELYON17	PCam	0.528	0.231	0.799	0.252

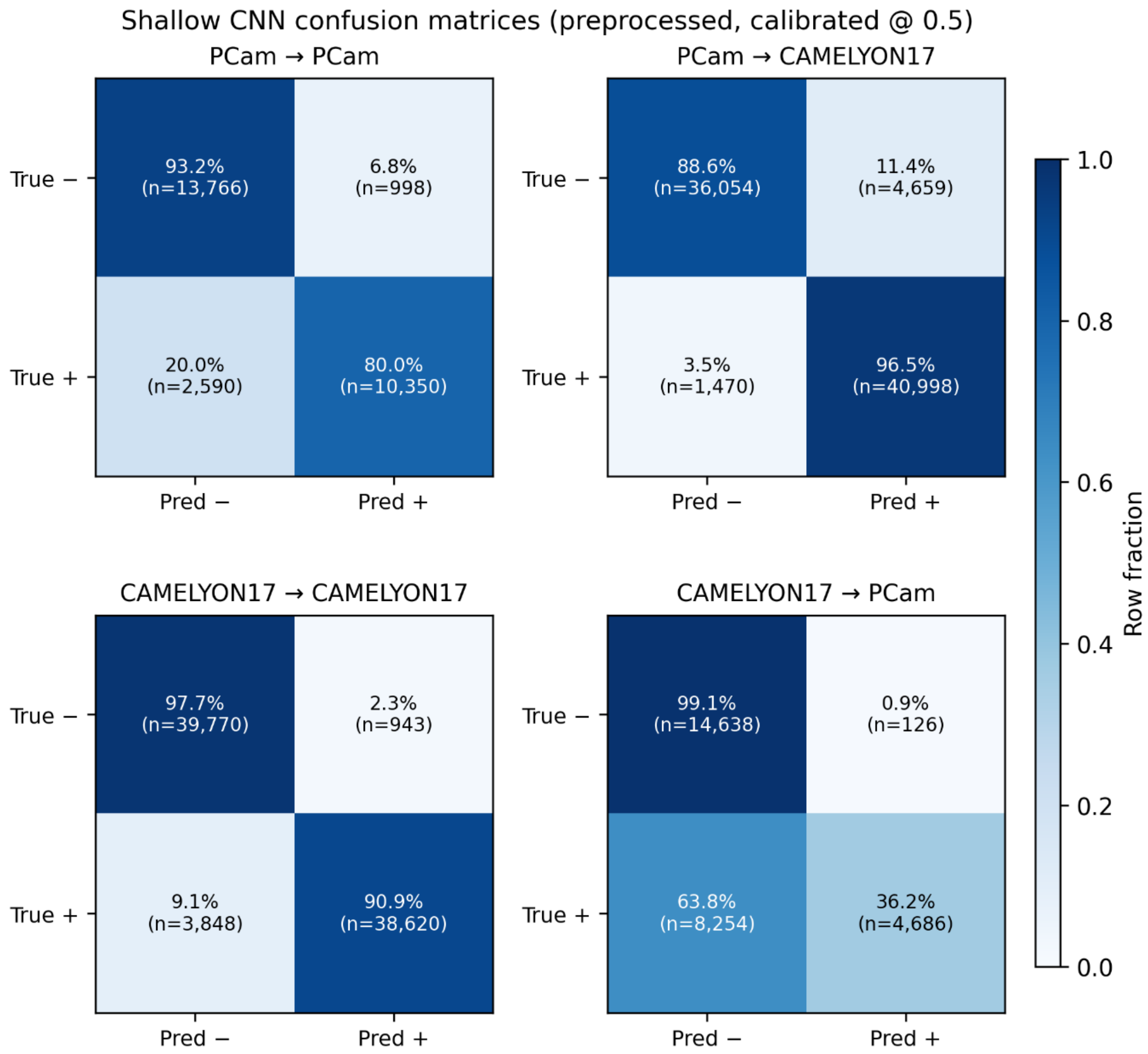

Figure 5.6. Shallow CNN confusion matrices on preprocessed corpora (calibrated, threshold 0.5).

Table 5.11 Shallow CNN transfer contrast on study-preprocessed patches (in-domain minus external).

Training arm	Δ ROC-AUC	Δ PR-AUC	Interpretation
PCam preprocessed	−0.037	−0.042	Higher ranking on external CAMELYON17 than in-domain PCam
CAMELYON17 preprocessed	+0.129	+0.121	Large drop on external PCam test

5.7.3 Preprocessing Pipeline Impact

Figure 5.7 compares raw and preprocessed ROC-AUC and recall on the same four train → test directions. Table 5.12 summarizes the change from raw to preprocessed on calibrated metrics at threshold 0.5. Deltas are computed as preprocessed minus raw on the same train → test direction; raw and preprocessed rows use different test counts where quality filtering removed tiles, so accuracy shifts on CAMELYON17 test partly reflect corpus composition as well as model behaviour.

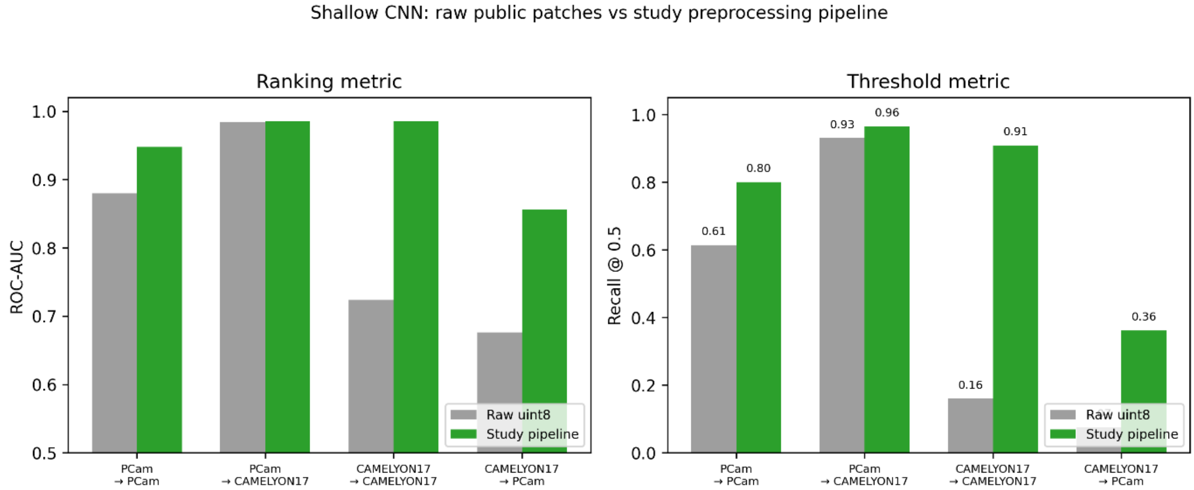

Figure 5.7. Shallow CNN: test ROC-AUC (left) and recall @ 0.5 (right) on raw uint8 patches (grey) versus the study pipeline (green), for four train → test directions; largest gain on CAMELYON17-trained, in-domain CAMELYON17 test.

Table 5.12 Change in shallow CNN performance when moving from raw inputs to the study preprocessing pipeline.

Train → Test	**ΔROC-AUC**	**ΔAccuracy**	**ΔRecall**	**ΔBrier**	**ΔECE_{15}**
PCam → PCam	+0.068	+0.091	+0.186	−0.067	−0.096
PCam → CAMELYON17	+0.001	−0.010	+0.04	+0.007	+0.045
CAMELYON17→CAMELYON17	+0.261	+0.388	+0.771	−0.223	−0.203
CAMELYON17 → PCam	+0.180	+0.177	+0.287	−0.077	−0.028

The pipeline had the largest effect on CAMELYON17-trained weights. On in-domain CAMELYON17 test, ROC-AUC rose from 0.724 to 0.985 and recall from 0.16 to 0.91 (Table 5.8 vs Table 5.9 and Table 5.10), so the raw arm was effectively unable to detect positives at a fixed 0.5 threshold despite moderate ranking (ROC-AUC 0.72). After preprocessing, the same architecture reached performance in line with the other strong cells in Table 5.9 and Table 5.10. PCam-trained in-domain performance also improved (ΔROC-AUC$= +0.068$, $\Delta recall = +0.19$, $\Delta Brier = -0.067$), showing that quality control, stain alignment, and value scaling mattered on the source domain and not only on the external hospital set.

Training dynamics mirrored test gains: best validation ROC-AUC increased from 0.931 to 0.947 for PCam arms and from 0.724 to 0.985 for CAMELYON17 arms. Raw CAMELYON17 training required a much larger post-hoc temperature ($T \approx 4.4$ versus 1.7 after preprocessing), indicating poorly structured raw logits even after calibration. Preprocessing did not remove transfer asymmetry. Table 5.11 still shows a 0.13-point ROC-AUC drop when CAMELYON17-trained weights are applied to external PCam, but it raised the floor of both in-domain and cross-domain performance relative to raw inputs.

The PCam → CAMELYON17 direction was already strong on raw external test (ROC-AUC 0.984) and changed little after preprocessing (ΔROC-AUC$= +0.001$), so stain-normalized tiles were less critical for that transfer direction than for learning usable CAMELYON17-native representations. All subsequent main experiments (Virchow2, bootstrap inference, and qualitative review) therefore use study-pipeline tensors only.

5.8 Virchow2 Classifier on Preprocessed Patches

Virchow2 was used with a frozen pretrained backbone and a trainable linear head (dropout 0.2). Four conditions (C1–C4) follow the same transfer layout: train on PCam or CAMELYON17, evaluate on the in-domain test split and on the external test split with the same study preprocessing pipeline. C1 and C2 (PCam-trained; in-domain and CAMELYON17 external) carry the headline external-validation question; C3 and C4 complete the prespecified mirror for asymmetry and threshold behaviour and are read alongside C1–C2, not as an equal success criterion for the forward path. Temperature was fit once per training run and applied to all test logits from that run.

Table 5.13 and Table 5.14 report calibrated test metrics. Table 5.15 gives raw sigmoid metrics on the same patches. Temperature scaling is a strictly monotone map of logits, so ROC-AUC and PR-AUC are identical between tables; Brier score, log loss, and ECE_{15} are not. The calibration step still matters for probability-based reporting: on C1, log loss fell from 0.175 to 0.170 and ECE_{15} from 0.027 to 0.020; on C4, where transfer stress was highest, log loss improved from 0.658 to 0.472 and ECE_{15} from 0.167 to 0.160, with a small Brier reduction (0.149 to 0.136). Accuracy at 0.5 was unchanged in these runs because the same threshold was applied after scaling, but the mapped probabilities were better aligned with empirical frequencies, supporting the post-hoc calibration step in the Methodology chapter rather than reporting raw sigmoid scores alone. Figure 5.8 shows confusion matrices for C1–C4 at threshold 0.5. PCam-trained runs share $T = 1.278$; CAMELYON17-trained runs share $T = 1.590$.

Table 5.13 Virchow2 test performance (ROC-AUC, PR-AUC, accuracy, balanced accuracy, precision) after temperature scaling (conditions C1–C4).

ID	Trained on	Tested on	ROC-AUC	PR-AUC	Accuracy	Balanced Acc.	Precision
C1	PCam	PCam	0.981	0.982	0.937	0.934	0.970
C2	PCam	CAMELYON17	0.997	0.998	0.984	0.984	0.987
C3	CAMELYON17	CAMELYON17	0.997	0.997	0.980	0.980	0.979
C4	CAMELYON17	PCam	0.950	0.953	0.824	0.812	0.988

Table 5.14 Virchow2 test performance (recall, F1, ntest, T, Brier, log loss, ECE15) after temperature scaling (conditions C1–C4).

ID	Trained on	Tested on	Recall	F1	n_{test}	T	Brier	Log Loss	ECE_{15}
C1	PCam	PCam	0.892	0.929	27,704	1.278	0.049	0.170	0.020
C2	PCam	CAMELYON17	0.982	0.984	83,181	1.278	0.013	0.056	0.010
C3	CAMELYON17	CAMELYON17	0.981	0.980	83,181	1.590	0.017	0.071	0.021
C4	CAMELYON17	PCam	0.630	0.770	27,704	1.590	0.136	0.472	0.160

Table 5.15 Virchow2 test performance on raw sigmoid outputs (conditions C1–C4).

ID	ROC-AUC	PR-AUC	Accuracy	Brier	Log loss	ECE15
C1	0.981	0.982	0.937	0.049	0.175	0.027
C2	0.997	0.998	0.984	0.013	0.053	0.003
C3	0.997	0.997	0.980	0.016	0.065	0.008
C4	0.950	0.953	0.824	0.149	0.658	0.167

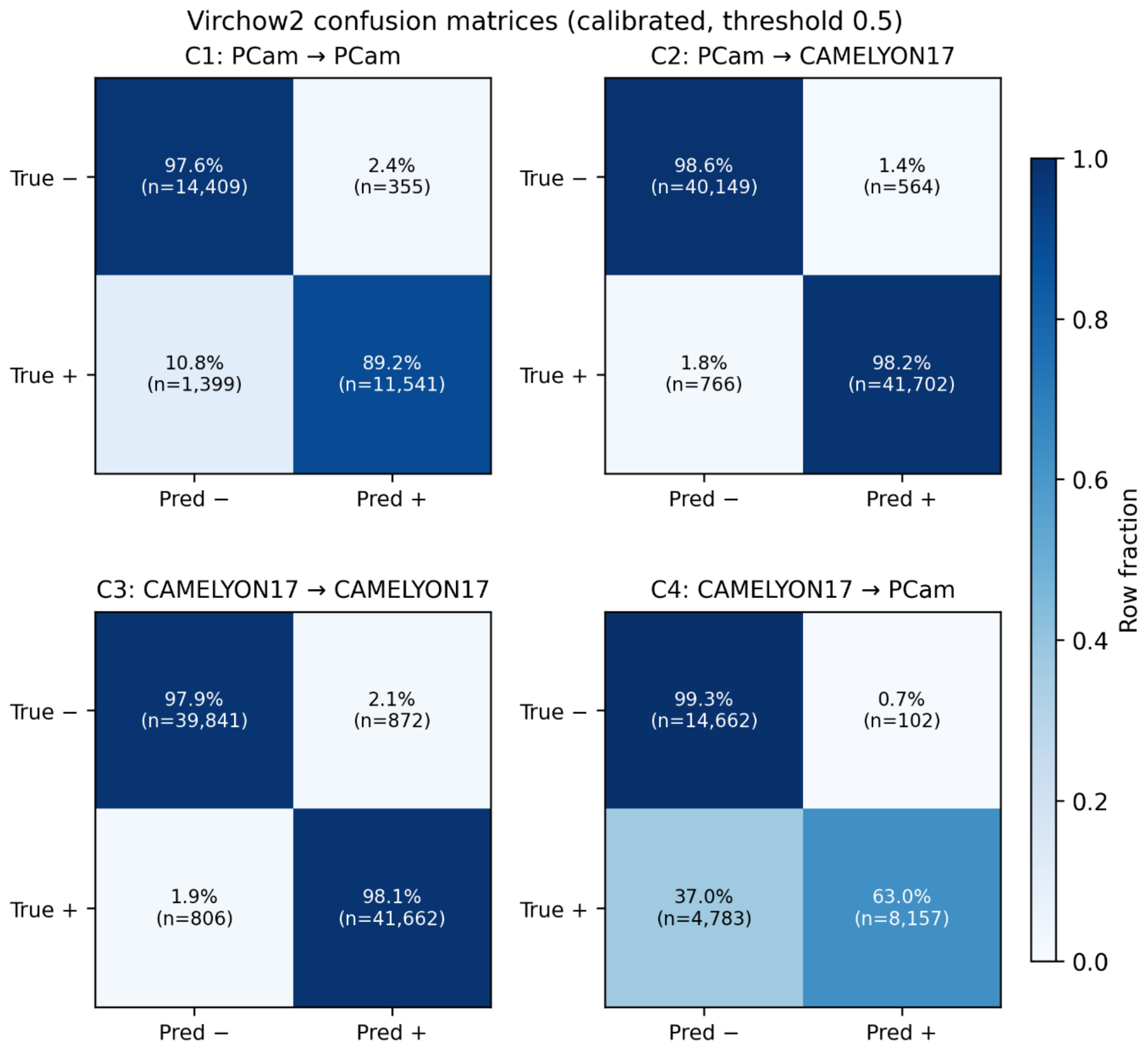

Figure 5.8. Virchow2 confusion matrices (calibrated, threshold 0.5; C1–C4).

Table 5.16 summarizes marginal cross-domain changes on calibrated metrics (external minus in-domain test surfaces of different sizes). The PCam-trained model ranked higher on CAMELYON17 test than on PCam test by ROC-AUC. The CAMELYON17-trained model lost 0.046 ROC-AUC points and 0.156 accuracy points on external PCam test.

Table 5.16 Virchow2 cross-domain change in ROC-AUC and accuracy (external minus in-domain).

Training origin	Δ ROC-AUC (ext. − in-domain)	Δ Accuracy (ext. − in-domain)
PCam (C2 − C1)	+0.016	+0.047
CAMELYON17 (C4 − C3)	−0.046	−0.156

C2 exceeded C1 on ROC-AUC and accuracy despite using the same PCam-trained weights. This pattern should be read cautiously: both scores were already near ceiling on PCam test (C1), the external CAMELYON17 test set is larger and has a slightly higher positive fraction, and ranking metrics can move upward when prevalence and score separability shift even if the underlying decision boundary is unchanged. It does not imply that unseen CAMELYON17 data are strictly easier for every patch type; it shows that PCam-trained Virchow2 transferred well to the external hospital set under the study pipeline.

Figure 5.9 shows reliability diagrams for C1 and C4 from the fifteen-bin expected calibration error analysis. Calibration was close on in-domain PCam test ($ECE_{15} = 0.020$) but degraded on external PCam test after CAMELYON17 training ($ECE_{15} = 0.160$), consistent with high precision and reduced recall in Table 5.13 and Table 5.14.

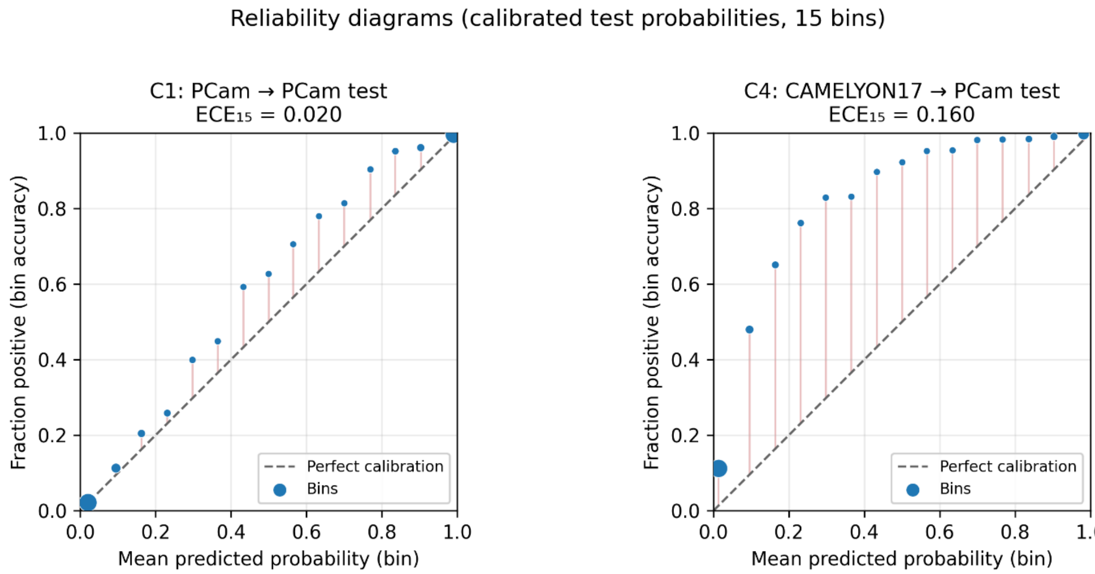

Figure 5.9. Reliability diagrams for Virchow2 calibrated test probabilities (15 bins): C1 on in-domain PCam test ($ECE_{15} = 0.020$) and C4 on external PCam test after CAMELYON17 training ($ECE_{15} = 0.160$). Points show mean predicted probability versus observed positive

5.9 Virchow2 versus Shallow CNN on Preprocessed Patches

Table 5.10 and Table 5.14 report the same transfer layout on identical post-QC test splits (n_{test} = 27,704 for PCam, 83,181 for CAMELYON17): a four-block convolutional baseline with a trainable head versus a frozen Virchow2 encoder with a linear head (dropout 0.2). Both families used the study preprocessing pipeline from Section 4.3, validation-selected checkpoints, and source-validation temperature scaling before test evaluation at threshold 0.5. Figure 5.10 shows test ROC-AUC on the four train → test cells for both models. Table 5.17 summarizes Virchow2 minus shallow CNN (positive Δ favours Virchow2 on ROC-AUC, accuracy, and recall; negative Δ favours Virchow2 on Brier and ECE). Table 5.18 and Figure 5.11 give cross-domain ROC-AUC change (external minus in-domain test) by training origin.

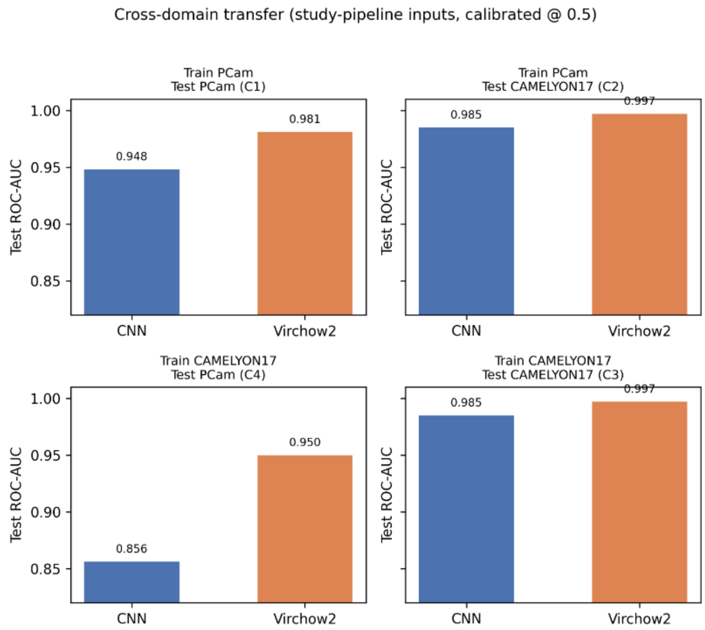

Figure 5.10. Cross-domain transfer on study-pipeline test splits: test ROC-AUC for shallow CNN and Virchow2 in each train → test cell (C1–C4 layout).

Table 5.17 Virchow2 minus shallow CNN on study-pipeline test metrics (calibrated @ 0.5).

Train → Test	**Δ ROC-AUC**	**Δ Accuracy**	**Δ Recall**	**Δ Brier**	**Δ ECE_{15}**
PCam → PCam	+0.033	+0.066	+0.092	−0.045	−0.023
PCam → CAMELYON17	+0.012	+0.058	+0.017	−0.041	−0.048
CAMELYON17→CAMELYON17	+0.012	+0.038	+0.072	−0.028	−0.006
CAMELYON17 → PCam	+0.094	+0.126	+0.268	−0.095	−0.092

Table 5.18 Cross-domain ROC-AUC change on study-pipeline inputs (external minus in-domain).

Training origin	**Virchow2**	**Shallow CNN**
PCam	+0.016	+0.037
CAMELYON17	−0.046	−0.129

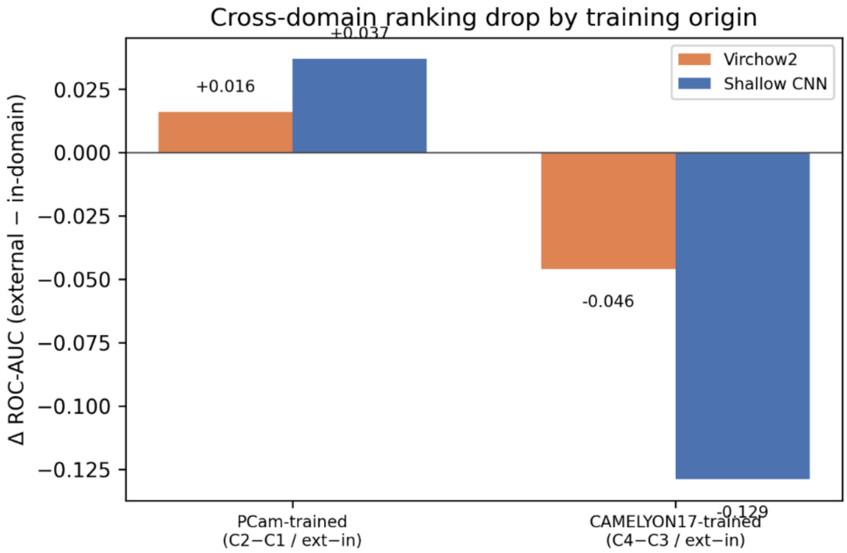

Figure 5.11. Cross-domain ROC-AUC change (external minus in-domain test) by training origin: Virchow2 vs shallow CNN.

Foundation-model features improved every preprocessed cell in Table 5.17. On in-domain PCam test, Virchow2 gained 0.033 ROC-AUC points and 0.066 accuracy points while roughly halving Brier score (0.049 versus 0.094). Gains on CAMELYON17 surfaces were smaller in ranking terms (+0.012 ROC-AUC in both directions where PCam or CAMELYON17 training matched the external hospital set) but still visible in accuracy and calibration. The largest separation appeared on the hardest transfer cell, CAMELYON17-trained weights evaluated on external PCam test (C4): Virchow2 retained ROC-AUC 0.950 and accuracy 0.824 versus 0.856 and 0.698 for the shallow CNN, with recall 0.630 versus 0.362 at similar precision (0.988 versus 0.974). At threshold 0.5, the CNN produced 8,254 false negatives on PCam test in that condition compared with 4,783 for Virchow2 (Figure 5.6 and Figure 5.8).

The two architectures followed the same qualitative transfer pattern but with different severity. Both ranked higher on external CAMELYON17 than on in-domain PCam when trained on PCam (Figure 5.11, positive Δ for PCam origin). Both lost performance on external PCam when trained on CAMELYON17; the CNN's in-domain-minus-external ROC-AUC gap was 0.129 points, about 2.8 times the Virchow2 gap of 0.046. Thus, preprocessing and representation capacity address different bottlenecks: the pipeline made CAMELYON17 learnable for a shallow CNN, while Virchow2 further reduced transfer loss and threshold failures on the difficult CAMELYON17 → PCam direction.

5.10 Bootstrap and permutation inference

Case-level bootstrap with $B = 2000$ replicates (seed 42) produced 95% percentile intervals for Virchow2 test metrics on calibrated probabilities. Table 5.19 lists primary intervals. Narrow bands on large CAMELYON17 test sets reflect stable ranking estimates.

Table 5.19 Virchow2 bootstrap 95% intervals on calibrated test metrics (point [2.5th, 97.5th percentile]).

ID	ROC-AUC	Accuracy	Brier	ECE_{15}
C1	0.981 [0.980, 0.983]	0.937 [0.934, 0.940]	0.049 [0.047, 0.050]	0.020 [0.018, 0.023]
C2	0.997 [0.997, 0.997]	0.984 [0.983, 0.985]	0.013 [0.013, 0.014]	0.010 [0.009, 0.011]
C3	0.997 [0.996, 0.997]	0.980 [0.979, 0.981]	0.017 [0.017, 0.018]	0.021 [0.020, 0.022]
C4	0.950 [0.948, 0.953]	0.824 [0.819, 0.828]	0.136 [0.133, 0.140]	0.160 [0.156, 0.164]

Paired bootstrap on the same test patches compared in-domain and externally trained weights on a shared test domain. Table 5.20 gives $\Delta = M_{in} - M_{ext}$ for PCam test (C1 minus C4) and CAMELYON17 test (C3 minus C2). All listed intervals excluded zero.

Table 5.20 Paired bootstrap transfer contrasts on shared test domains.

Test domain	Contrast	Δ ROC-AUC [95% CI]	Δ Accuracy [95% CI]	Δ Brier [95% CI]
PCam	C1 − C4	0.031 [0.029, 0.033]	0.113 [0.109, 0.117]	−0.088 [−0.090, −0.085]
CAMELYON17	C3 − C2	−0.0005 [−0.0008, −0.0002]	−0.0042 [−0.0051, −0.0033]	+0.0039 [0.0035, 0.0045]

Paired permutation tests (10,000 sign flips) on per-patch Brier and log-loss contributions compared training origins on the same test surface. Table 5.21 reports confirmatory tests with Benjamini–Hochberg adjustment at $q = 0.05$. All four comparisons rejected the null of equal mean patch-wise scores.

Table 5.21 Paired permutation tests on shared test patches.

Test surface	Score	Comparison	p	pBH
PCam test	Brier	C1 vs C4	$< 10^{-4}$	$< 10^{-4}$

PCam test	Log loss	C1 vs C4	$< 10^{-4}$	$< 10^{-4}$
CAMELYON17 test	Brier	C3 vs C2	$< 10^{-4}$	$< 10^{-4}$
CAMELYON17 test	Log loss	C3 vs C2	$< 10^{-4}$	$< 10^{-4}$

5.11 Qualitative error analysis

Human checklist review was completed for all four Virchow2 conditions. Up to 40 patches per transfer direction were exported across four buckets (false positive, false negative, high-entropy error, high-entropy correct), with an optional fifth bucket adding up to ten more high-confidence errors. Figure 5.12 links the full-test error pools to the numeric results in Section 5.9: C4 stands out for 4,783 false negatives available versus only 102 false positives, consistent with the recall drop in Table 5.14. Table 5.22 summarizes checklist marks in the false-negative and high-entropy error buckets only $(k/10)$ present in the reviewed sample of ten patches per bucket per condition; not full-test prevalences.

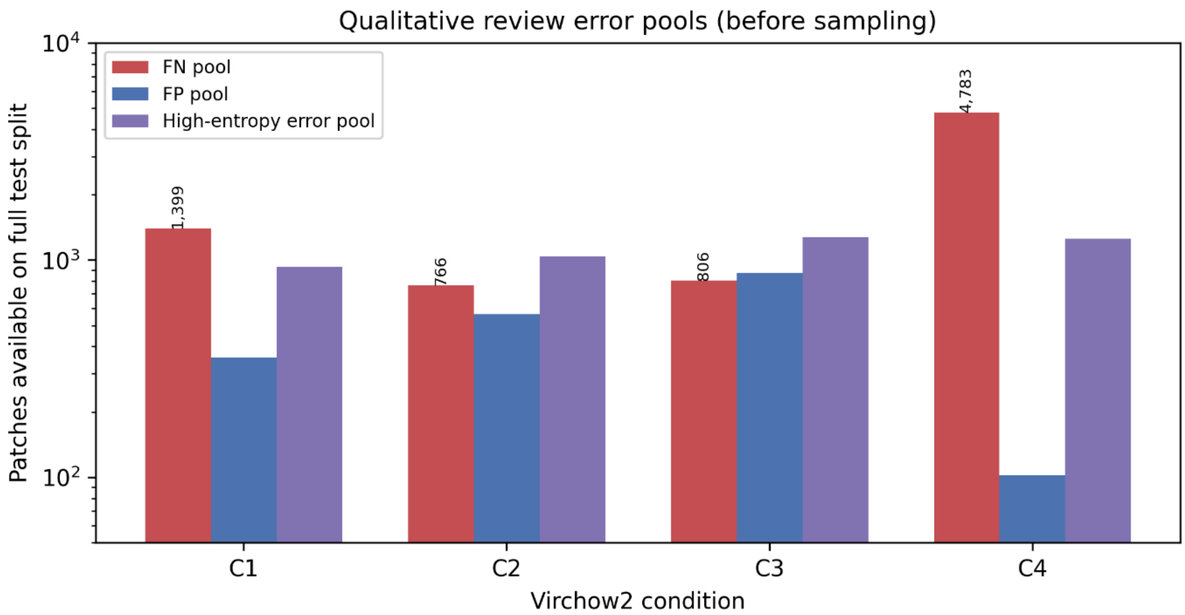

Figure 5.12. Virchow2 qualitative-review error pools on full test splits.

Table 5.22 Qualitative checklist marks in false-negative and high-entropy error buckets (reviewed sample, k of 10).

Pattern	C1 FN	C2 FN	C3 FN	C4 FN	C1 HE Err	C2 HE Err	C3 HE Err	C4 HE Err
Patch-context limitation	8	6	4	6	4	4	4	5

Small-focus lesion	5	4	4	6	4	4	5	4
Colour/stain atypia	2	6	2	4	2	4	2	4
Borderline morphology	4	4	4	4	7	8	7	8

On C1, reviewed false negatives often carried patch-context limitation (8/10), matching the PCam central-focus label rule despite strong overall metrics. On C2, colour/stain atypia was common in reviewed false negatives (6/10) and high-entropy errors (4/10). even though transfer to CAMELYON17 test remained excellent numerically. C3 showed no single dominant checklist pattern among false negatives, in line with strong in-domain performance. C4 combined the largest false-negative pool with reviewed FN cases frequently marked for small-focus lesion and context limits (6/10 each), aligning with the threshold and calibration failures in Table 5.13 and Table 5.14. Across all conditions, borderline morphology appeared in 7–8/10 high-entropy errors, suggesting uncertain wrong predictions are driven more by ambiguous morphology than by gross tissue loss or artifacts alone. These patterns support the quantitative transfer story in Sections 5.8-5.10 but do not replace it.

5.12 Interpretation

The experiments trace a single pipeline from public patch benchmarks through shared quality control and Macenko stain normalization to two classifier families evaluated under strict transfer rules. The pipeline was designed so that a classifier trained on the large PCam development corpus could be stress-tested on held-out multi-hospital CAMELYON17 test; the data stages clarify what was removed and why, while the modelling stages separate representation capacity from preprocessing and calibration choices on that path and on the prespecified mirror.

The headline empirical result is strong forward external transfer: after the study pipeline, PCam-trained Virchow2 reached very high discrimination and well-behaved probabilities on CAMELYON17 test (C2), with a clear margin over the shallow CNN on the same tensors, which is evidence that integrity-aware curation plus a frozen foundation encoder can support deployment-style generalization from a curated tile benchmark to a harder external nodal benchmark under fixed reporting rules. That claim is intentionally scoped to the direction tested here; it does not assert uniform safety on every future centre or slide-level workflow without further validation.

The mirrored direction (CAMELYON17 train, PCam test) shows how asymmetric transfer can be: both architectures retained respectable ROC-AUC on external PCam, yet Virchow2 and especially the CNN lost threshold recall and calibration relative to in-domain use (C4), with a large false-negative pool despite high precision at 0.5. Read together, C2 and C4 separate ranking strength from operating-point and probability behaviour when the training domain and label geometry differ from the test surface, which is why the study reports calibration and confusion structure alongside area metrics.

Integrity checks showed that duplicate and cross-dataset overlap were negligible at the byte level. Quality filtering removed a modest fraction of tiles, mostly for low tissue, without shifting class balance. The stain benchmark justified Macenko for full-corpus work even though another arm achieved slightly better threshold accuracy on a smaller balanced subset. That distinction matters when reading later results: ranking metrics and fixed-threshold metrics can favour different preprocessing strategies on a shallow network, but Macenko remained the prespecified choice for all main runs.

The shallow CNN results establish how much performance depends on preprocessing and on which transfer leg is read. Without the study's preprocessing pipeline, CAMELYON17-trained weights showed weak in-domain performance on raw CAMELYON17 test (ROC-AUC 0.72 with recall 0.16 at 0.5), which confirms that raw public patches alone were not a fair match to the curated corpora used elsewhere. After the full pipeline, the CNN reached a strong ranking on CAMELYON17 when trained on PCam, in line with Virchow2 on that forward path. The mirror leg remained harder: CAMELYON17-trained weights on PCam test showed larger drops in accuracy and recall and worse calibration than the forward leg, with the CNN suffering a larger ROC-AUC gap than Virchow2.

Virchow2 improved every preprocessed comparison in Table 5.17 by a meaningful margin. The gain was not limited to the in-domain PCam test; it persisted on external surfaces and was largest when CAMELYON17-trained weights were applied to PCam—the mirror stress cell—where Virchow2 preserved substantially higher recall at similar precision than the CNN. Bootstrap intervals were tight on large test sets but still excluded trivial shifts for paired contrasts on shared patches. Permutation tests on patch-wise Brier and log-loss confirmed that the two training origins produced different probability behaviour on the same tiles, not only different aggregate thresholds.

Structured checklist review (Section 5.11) aligned with these numeric patterns: context-limited and small-focus false negatives on external PCam (C4), stain atypia on cross-domain CAMELYON17 test (C2), and borderline morphology among high-entropy errors in every condition.

Taken together, the results support using this pipeline and the PCam-trained Virchow2 configuration as a reusable base for further external corpora and sites and for closer clinical-style validation (slide-level aggregation, operating-point choice, prospective checks), while keeping claims proportional to patch benchmarks and fixed splits. On the mirror leg, a deployment that relied on a fixed 0.5 cutoff without target-domain tuning would still be risky despite strong ROC-AUC; future work can target recall there through class weighting, focal losses, or threshold selection on validation material from the intended deployment domain, and can extend the qualitative protocol with formal inter-rater agreement on a larger stratified sample.

Chapter 6 Limitations and Future Work

This section delineates the boundaries of what the current empirical evidence can definitively claim by design, outlines the structural scope of the study, and establishes clear trajectories for subsequent research.

6.1 Limitations

By design, this study is retrospective and relies exclusively on public, patch-level benchmarks. Consequently, the findings characterize computational discrimination, probability calibration, and error behaviour under fixed dataset splits and established label definitions; they do not evaluate clinical workflow integration, regulatory approval metrics, or patient outcomes. Furthermore, the binary task itself (classifying 96×96 pixel patches based on tumour presence within a central 32×32 focus) serves as a highly controlled surrogate for whole-slide analysis. It does not fully encapsulate the complexity of full whole-slide search routines, isolated tumour cell (ITC) classifications, or the multidimensional clinical costs associated with practical hospital deployment.

A major constraint governing the practical scope of this study was the finite GPU capacity, wall-clock limits on full-corpus training, and the substantial computational overhead required to process massive gigapixel-scale image matrices end-to-end. This hardware envelope bounded the experimental matrix to a single reproducible pipeline. Investigating additional colour normalizers on full data, testing alternative foundation backbones, conducting exhaustive hyperparameter or threshold sweeps, or executing full leave-one-hospital-out validation blocks would have multiplied training and inference passes beyond our available runtimes.

Statistical uncertainty estimates, including bootstrap percentile intervals and paired permutation tests, quantify variability strictly within these held-out tensors rather than substituting for validation on an independent, prospectively gathered clinical cohort. Similarly, the structured qualitative review provides a targeted checklist evaluation of small, randomly drawn error and high-entropy pools. It functions as an illustrative diagnostic tool for visual patterns (such as borderline morphology or patch-context boundaries) rather than a complete, full-test census estimating true absolute prevalence across the entire dataset.

6.2 Future Work

The empirical patterns observed across these experiments suggest several natural extensions to advance the framework. The highest-priority line of work is to stress-test the PCam-trained models (in particular the Virchow2 configuration that performed strongly on CAMELYON17 test) under progressively stricter evidence standards than patch benchmarks alone: further public or consented patch corpora from new institutions or scanners; hospital-stratified or leave-one-institution-out analysis on CAMELYON17 to probe multi-centre stability of the forward external result; richer perturbation or stress tests; and, where ethics and data access allow, prospective sampling or reader-in-the-loop studies with operating-point analysis tied to clinical sensitivity targets. Slide-level or weakly supervised evaluation on whole lymph-node slides, together with workflow metrics such as latency, escalation rules, and failure handling, would be needed before any responsible claim about broad generalisability or readiness for clinical deployment. The thesis should be read as establishing a strong patch-level forward transfer benchmark and a reusable evaluation protocol for that path, not as deployment clearance.

Further methodological work should expand the stain-handling investigation beyond the initial subset benchmark. While the single-reference Macenko method was utilised for full-corpus preprocessing, evaluating alternative classical or learned colour normalizers across the full datasets would test how much headroom remains on the forward PCam-to-CAMELYON17 external path after the current choice, and whether alternative normalizers materially change the mirror path (CAMELYON17-trained models on external PCam test), where ranking remained strong but recall at a fixed threshold and calibration were stressed.

Transitioning from a frozen-encoder framework to evaluating the effects of fine-tuning top transformer blocks or testing alternative classification heads represents a key step for model optimisation. Furthermore, because the standard 0.5 probability cutoff revealed distinct recall failures on the hardest transfer domain despite strong underlying ranking metrics (ROC-AUC = 0.950), subsequent investigations should integrate cost-sensitive thresholds, sensitivity targets at fixed false-positive rates, and prevalence-aware objectives directly into the evaluation pipeline, with the forward external target kept explicit when reporting deployment-style claims.

Addressing these domain shifts also requires exploring target-aware calibration strategies. Incorporating lightweight adapters or test-time probability scaling using small, unlabelled validation samples from the target clinical environment could bridge the reliability gap highlighted by the degraded out-of-domain reliability diagrams on the mirror leg, and could be studied alongside the forward PCam-trained model on CAMELYON17 test where calibration was already strong under this protocol.

Finally, the localised patch-level architecture developed here should ultimately be scaled up to slide-level clinical decision support systems. The strong forward transfer result on PCam to CAMELYON17 test under this pipeline motivates repeating the same integrity-aware recipe as new corpora become available. Merging foundation-model patch embeddings with advanced weakly supervised MIL aggregators, expanding labels to capture explicit size or diagnostic strata, and measuring operational deployment metrics, such as processing throughput and memory footprint, will be essential to translate these computational insights into trustworthy digital pathology workflows.

Chapter 7 Conclusion

This thesis establishes a reproducible computational framework to evaluate patch-level binary metastasis detection on PatchCamelyon (PCam) and CAMELYON17 (WILDS), with the principal external readout defined as performance when training on PCam and evaluating on held-out multi-hospital CAMELYON17 test. This stress case motivated integrity-aware curation. A prespecified mirrored layout (train on CAMELYON17, test on PCam) is reported in full so that transfer asymmetry, threshold behaviour, and calibration can be read transparently; it is not treated as a symmetric second success criterion for the forward claim. Driven by the clinical need to mitigate the latency and high inter-observer variability of manual sentinel lymph node staging, the work examines how data-integrity screening, Macenko stain normalization, and network capacity affect diagnostic robustness to site-specific distribution shifts. The empirical evidence demonstrates that explicit preprocessing and foundational representation capacity resolve entirely distinct bottlenecks. For a conventional convolutional network, raw public inputs proved operationally unlearnable, yielding a failing in-domain CAMELYON17 threshold recall of 0.16, whereas enforcing the study pipeline caused in-domain ROC-AUC to surge to 0.985 and recall to rise to 0.91, alongside substantial source-domain enhancements on PCam.

Concurrently, the frozen pretrained representations of Virchow2, with a trainable linear head, outperformed the shallow CNN baseline in every experimental cell, with the clearest practical margin on the forward PCam-to-CAMELYON17 external test. On that headline path, PCam-trained Virchow2 approached ceiling discrimination on CAMELYON17 test (ROC-AUC 0.997 and accuracy 0.984 at the fixed 0.5 threshold after calibration). The mirror direction is where high-stakes threshold failure appears: strong discriminative ranking on external PCam (ROC-AUC 0.950) coexisted with a severe collapse in fixed-threshold sensitivity (recall 0.630), missing 4,783 positive patches on the full test split. Validation-bound post-hoc temperature scaling improved probabilistic alignment but did not restore recall at 0.5, and out-of-domain reliability eroded, with expected calibration error (ECE_{15}) widening from 0.020 on in-domain PCam to 0.160 on external PCam after CAMELYON17 training. Case-level paired bootstrap percentile intervals and paired permutation tests confirmed that these shifts were highly significant under Benjamini–Hochberg multiplicity control.

These quantitative dynamics were reinforced by structured qualitative reviews, which mapped algorithmic failures to visible tissue and context factors. In-domain false negatives were frequently driven by patch-context limitations where metastatic cells fell outside the central-focus label rule, whereas transfer-stress errors concentrated around colour or stain atypia on external CAMELYON17 data and around small-focus tumour patterns on external PCam tissue. Crucially, borderline cellular morphology appeared in nearly 80% of high-entropy errors across all conditions, suggesting that algorithmic uncertainty maps tightly onto true biological ambiguity rather than random imaging artefacts.

Ultimately, this thesis demonstrates that a disciplined, integrity-first pipeline paired with a frozen pathology foundation model can deliver near-ceiling external transfer from PCam to multi-hospital CAMELYON17 test, which is precisely the stress case that motivated the work, while still documenting, through a prespecified mirror, how ranking strength and day-to-day operating-point behaviour can diverge. That combination is a constructive contribution: it gives future projects a reusable evaluation template, a strong positive benchmark outcome, and interpretable error biology tied to the models' mistakes. Carrying the same standards forward (headline performance on the intended external target, calibration and threshold review, and extension to richer corpora and slide-level settings when resources allow) is the natural next step from the solid foundation this dissertation establishes.

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